

TARGET-90

JKBOSE CLASS-11TH

CHEMICAL BONDING

and
MOLECULAR STRUCTURE.

One Shot

CHEMISTRY

Vishal Sir

Bhaichara
ON
TOP.



Topics To Be Covered

1 ... One Shot

2 ...

3 ...

4 ...

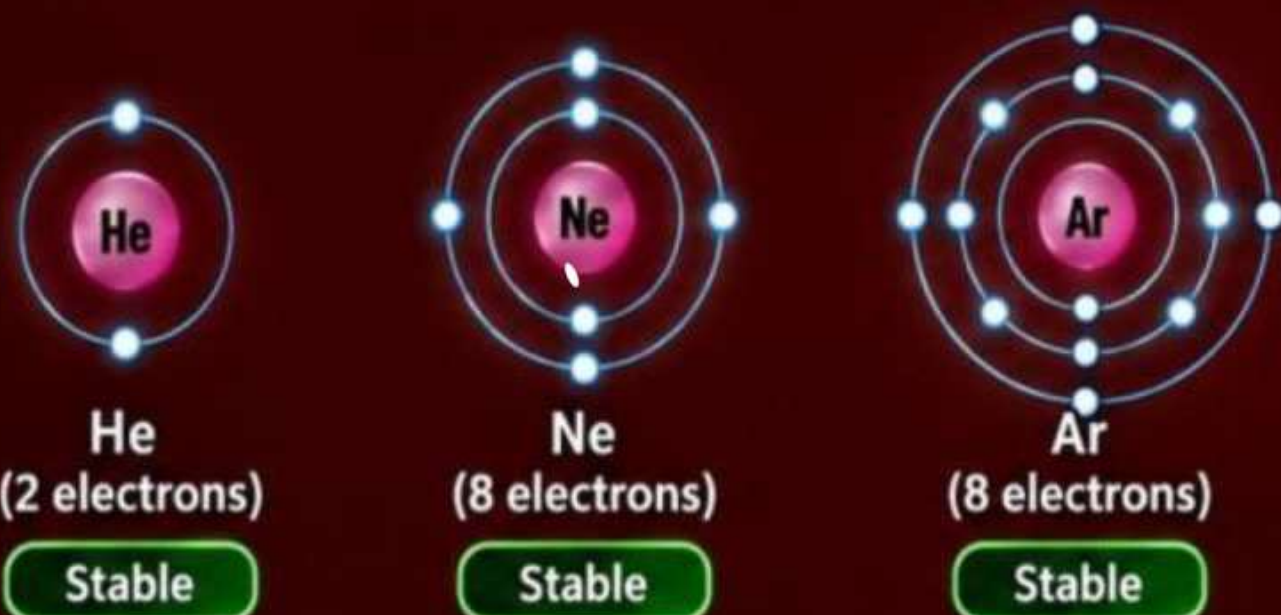
Kossel and Lewis Approach to Chemical Bonding



(Why Do Atoms Combine?)

1 To get noble gas configuration

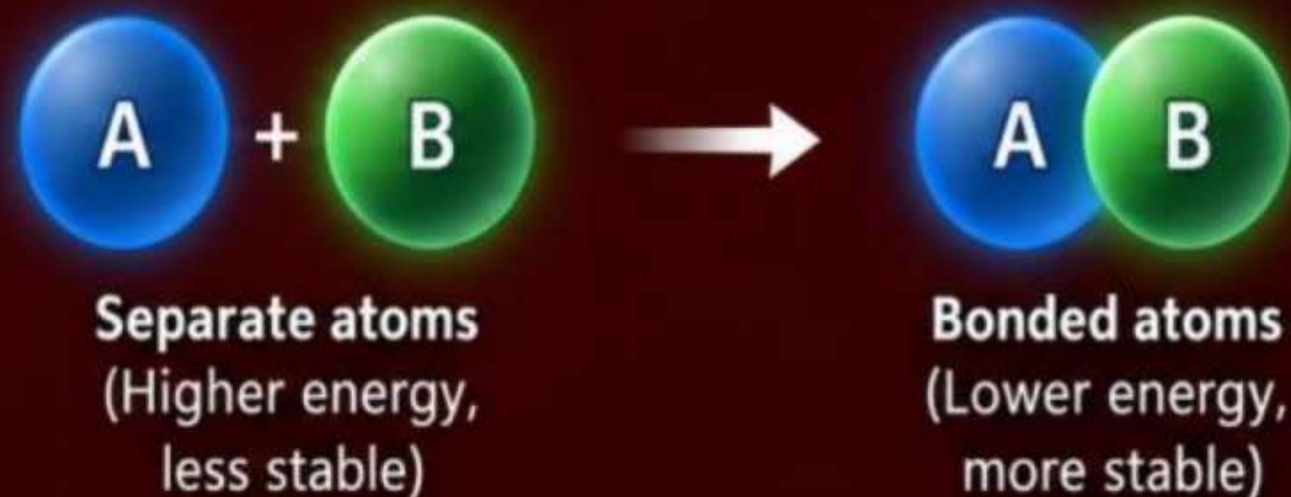
Noble gases are stable because they have **8 electrons** in their outermost shell (except He, which has 2). $(n^2 n^6)$



➔ This tendency of atoms to complete 8 electrons in their outermost shell for stability is called the **octet rule**.

2 To gain stability by lowering energy

When atoms combine, their **total energy decreases**, and they become more stable.



➔ If energy doesn't decrease, the bond **won't form**.





Lewis Dot Symbols



» G.N. Lewis gave simple symbols to show valence electrons — called **Lewis symbols** or **electron dot symbols**.

Lewis dot symbols (up to Calcium)

1 H • 1 electron	2 He : 2 electrons	3 Li • 1 electron	4 Be : 2 electrons	5 • B • 3 electrons	6 • C • 4 electrons	7 • N • 5 electrons	8 • O • 6 electrons	9 • F • 7 electrons	10 • Ne • 8 electrons
11 Na • 1 electron	12 Mg •• 2 electrons	13 Al • 3 electrons	14 • Si • 4 electrons	15 • P • 5 electrons	16 • S • 6 electrons	17 • Cl • 7 electrons	18 • Ar • 8 electrons	19 K • 1 electron	20 Ca •• 2 electrons

» Number of dots = number of **valence electrons**.



Ionic Bond (Transfer of electrons → Stability)



Definition

An **ionic bond** is formed when one atom **loses** electron(s) and another atom **gains** electron(s) to attain **noble gas configuration**.



Key Terms

- **Electrovalency:**
Number of electrons lost or gained by an atom.
- **Ionic compounds:**
Substances formed by ionic bonds.



Factors Affecting Ionic Bond Formation

1 Low Ionisation Energy

- Atom should easily lose electron(s).
- Lower ionisation energy → **easier to form cations** → ionic bond forms easily.

(Group 1 & Group 2)



2 High Electron Gain Enthalpy

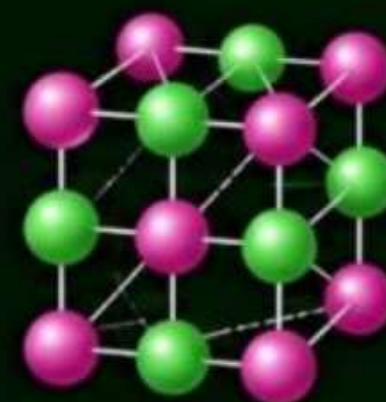
- Atom should easily gain electron(s).
- More negative electron gain enthalpy → **easier to form anions** → ionic bond forms easily.

Group 17



3 High Lattice Enthalpy

- Lattice enthalpy is the **enthalpy released** when one mole of an ionic solid is formed from its gaseous ions.
- Higher lattice enthalpy → stronger ion-ion attraction → **more stable** ionic compound.





Ionic Bond

(Transfer of electrons → Stability)



Examples of Ionic Bond Formation



Definition

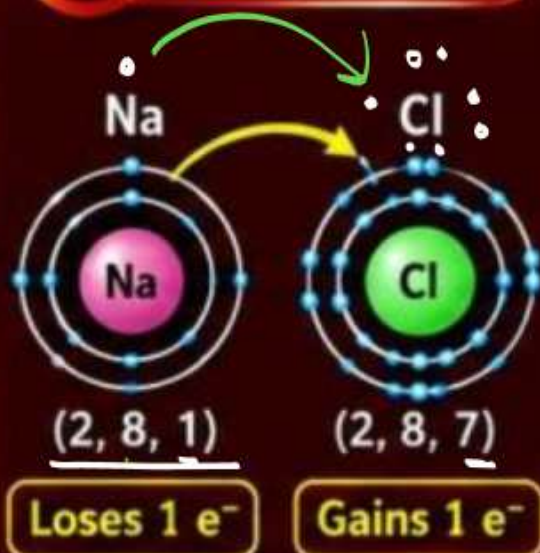
An **ionic bond** is formed when one atom **loses** electron(s) and another atom **gains** electron(s) to attain **noble gas configuration**.



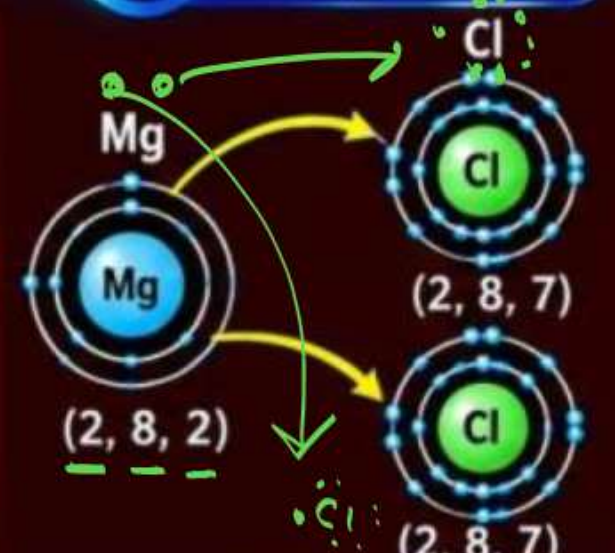
Key Terms

- **Electrovalency:**
Number of electrons lost or gained by an atom.
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Substances formed by ionic bonds.

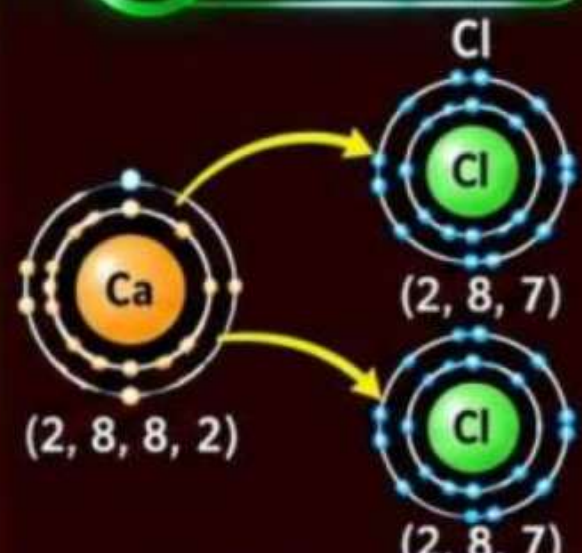
1 NaCl



2 MgCl₂



3 CaCl₂





Covalent Bond

Ionic Bond $\rightarrow$ Metal Non-metal
NaCl



Definition

A covalent bond is formed when two atoms **share electrons** to complete their outer shell (**octet or duplet**).
It usually forms between two non-metal atoms.
The number of electrons an atom shares is called its covalency.

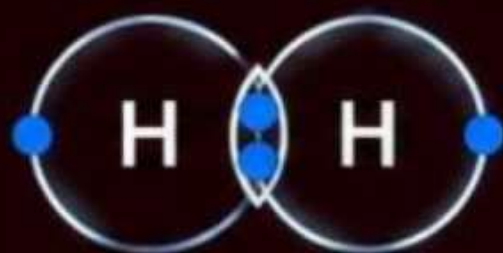


Types of Covalent Bonds

1

Single Covalent Bond

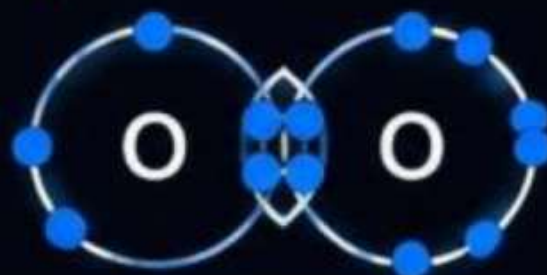
- Atoms share **one pair** of electrons.
- Example:** $\text{H}_2 \rightarrow \text{H}-\text{H}$ 



2

Double Covalent Bond

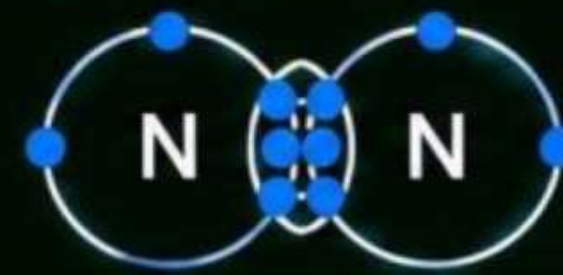
- Atoms share **two pairs** of electrons.
- Example:** $\text{O}_2 \rightarrow \text{O}=\text{O}$



3

Triple Covalent Bond

- Atoms share **three pairs** of electrons.
- Example:** $\text{N}_2 \rightarrow \text{N} \equiv \text{N}$



STRUCTURE WRITING LEWIS STRUCTURES OR ELECTRON DOT SYMBOLS OF VARIOUS MOLECULES



STEPS :

① Calculate total valence electrons (TVE).

$$TVE = \text{Sum of valence } e^- \text{ of all atoms} + (-) \text{ charge (add)} - (+) \text{ charge (subtract)}$$

② Selection of central atom.

- Atom with least electronegativity is taken as central atom.
- If there is a choice, atom present in least number is taken as central atom.
- H and F never act as central atom.

③ Total number of bonds.

$$\text{Required valence } e^- \text{ (for central atom)} = \frac{TVE}{2}$$

④ Distribute the remaining electrons so that each atom follows octet rule (except H).

Example: CO_2

1) Total valence electrons (TVE):

$$C = 4, \quad O = 6 \times 2 = 12$$

$$TVE = 4 + 12 = 16 e^-$$

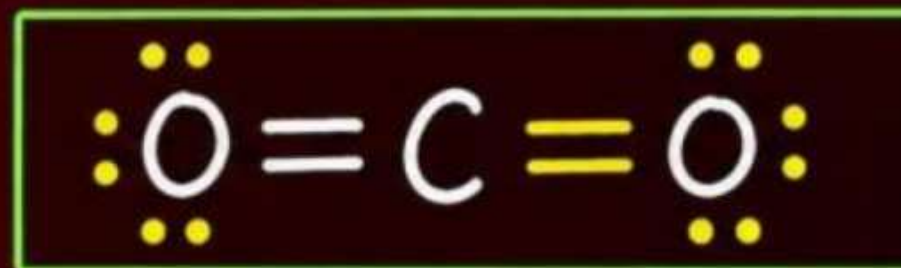
Valence e^-	
C	4
O	6

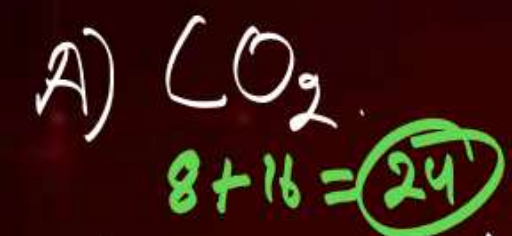
2) Central atom: C (less electronegative)

3) Total number of bonds:

$$\text{Required valence } e^- = \frac{TVE}{2} =$$

4) Arrange the atoms and distribute remaining electrons (octet rule):





$$\begin{aligned} \text{C} &= 14 \text{ group} = 4e^- \\ 2\text{O} &= 16 \times 2 = 32e^- \end{aligned}$$

1) $\text{TVE} = \text{Valence } e^- + (\text{ve charge}) - (\text{+ve charge})$
 $= 16 + 0 + 0$
 $= 16e^-$

2) Central atom $\rightarrow \text{C}$

3) Total no. of bond $\rightarrow \frac{\text{Reqd. V.E} - \text{TVE}}{2}$
 $= \frac{24 - 16}{2} = \frac{8}{2} = 4$

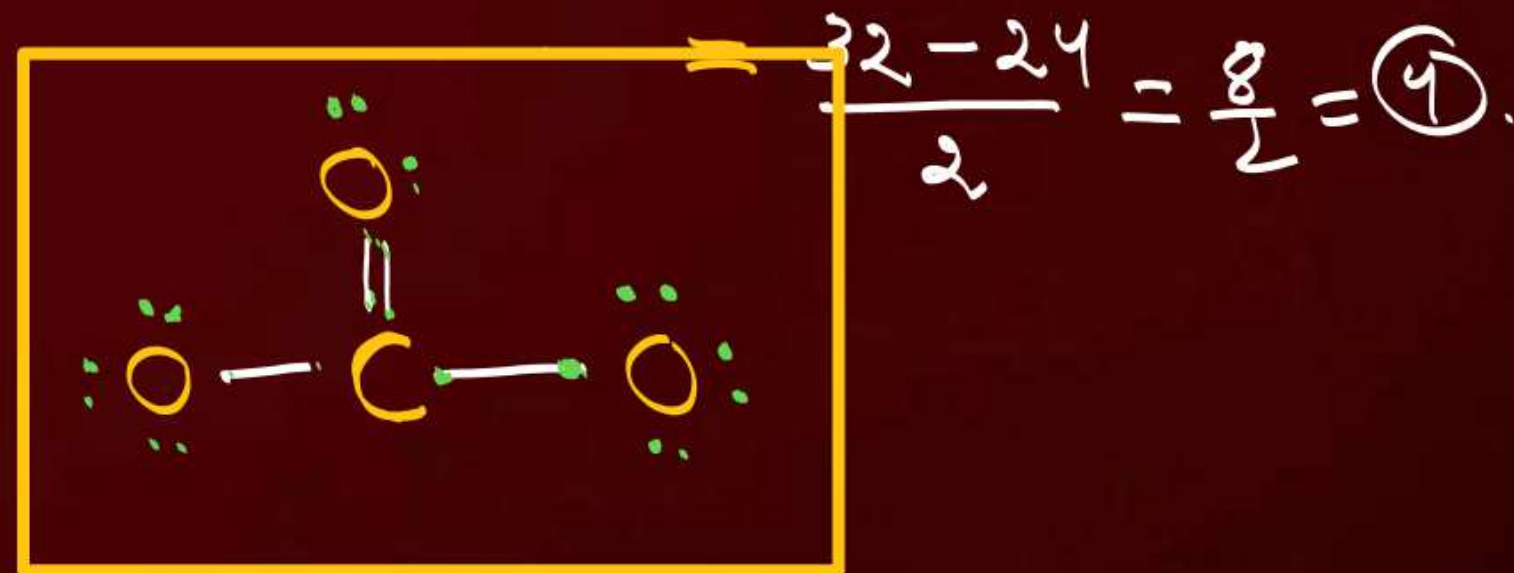


$$\begin{aligned} \text{C} &= 8 \\ 3 \times \text{O} &= 8 \times 3 = 24 \\ &= 32 \end{aligned}$$

1) $\text{TVE} = 4 + (6 \times 3) + 2$
 $= 4 + 18 + 2 = 24e^-$

2) Central atom = C

3) Total no. of bond $\rightarrow \frac{\text{Reqd. V.E} - \text{TVE}}{2}$



e.g. 1) H_2O 2) NH_3 3) SO_2 4) NO_2^- 5) NO_3^-

6) NH_4^+ 7) O_3

CALCULATION OF FORMAL CHARGE →

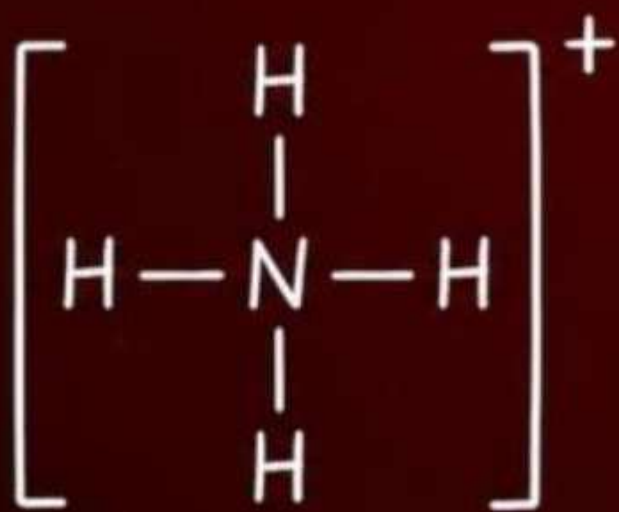
$$F.C. = V - L - \frac{1}{2} S$$

V = Total valence e^- of atom

L = Total e^- in lone pairs on atom

S = Total e^- in shared pairs (bonding e^-)

e.g. Find the formal charge on nitrogen in NH_4^+ .



$$V(N) = 5$$

$$L(N) = 0$$

$$S(N) = 8 \text{ (4 bonds)}$$

$$F.C.(N) = 5 - 0 - \frac{1}{2}(8)$$

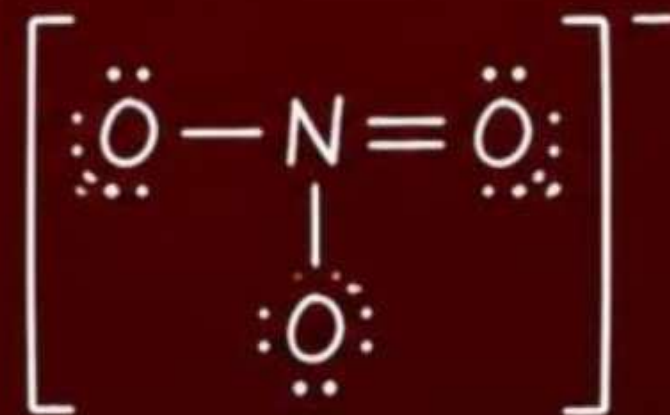
$$= 5 - 4 = +1$$

Steps :

- ① Draw the Lewis structure.
- ② Find V , L and S for the atom.
- ③ Apply $F.C. = V - L - \frac{1}{2} S$.
- ④ Calculate the formal charge.

Another example :

Find the formal charge on oxygen in NO_3^- .



For singly bonded O :

$$V = 6$$

$$L = 6$$

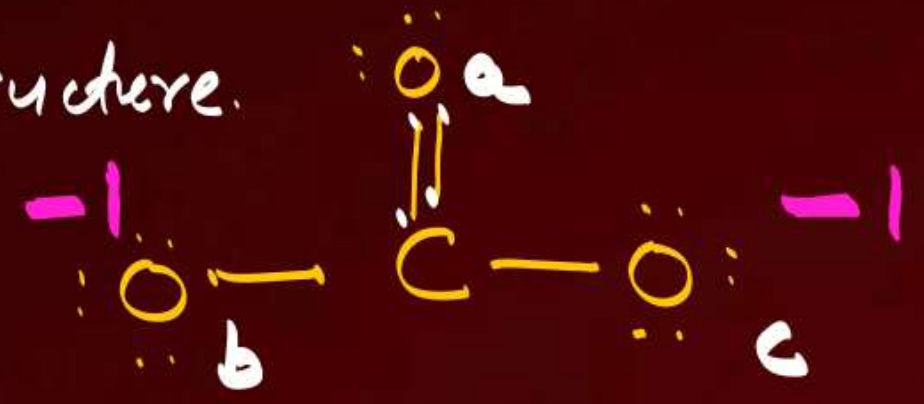
$$S = 2$$

$$F.C. = 6 - 6 - \frac{1}{2}(2)$$

$$= 6 - 6 - 1 = -1$$

Q. F.C calculation of CO_3^{2-}

① Lewis structure.



i) F.C on $\text{O}_a = V - L - \frac{1}{2} S$

$$= 6 - 4 - \frac{1}{2} \times 4$$

$$= 6 - 4 - 2$$

$$= 6 - 6$$

$$= 0$$

+ve charge $\rightarrow$ Central Atom

-ve charge = side atom

ii) F.C on $\text{O}_b = V - L - \frac{1}{2} S = 6 - 6 - \frac{1}{2} \times 2$

$$= 6 - 6 - 1$$

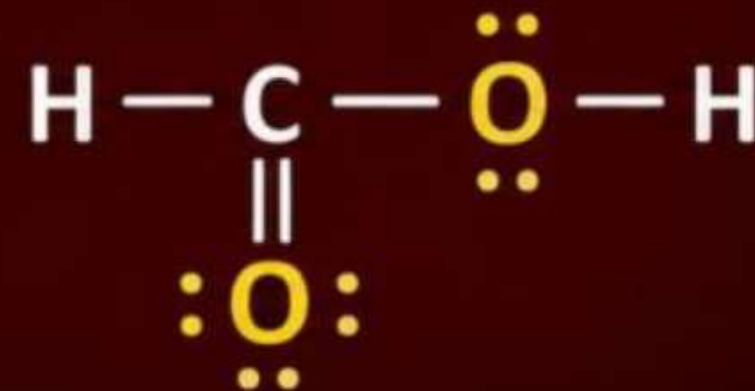
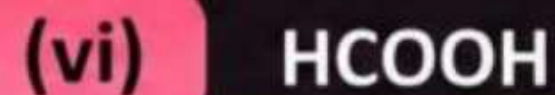
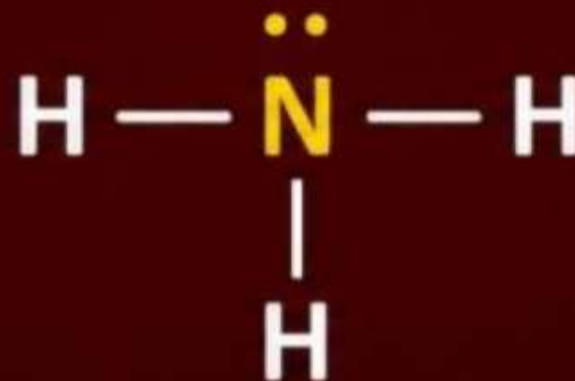
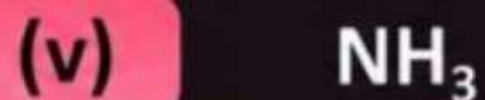
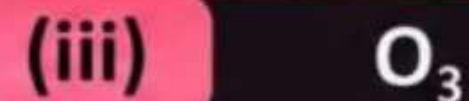
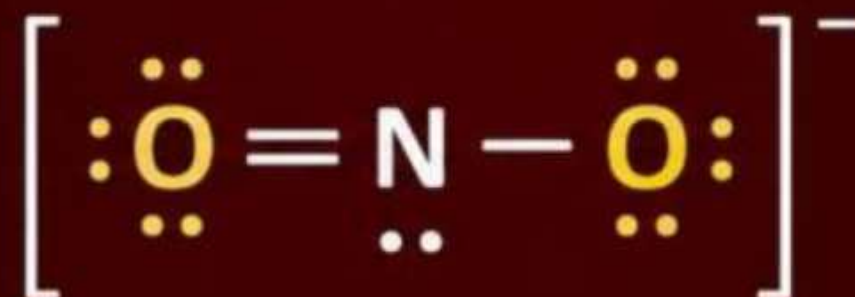
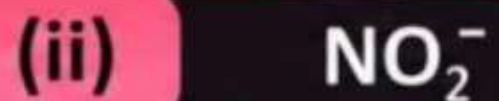
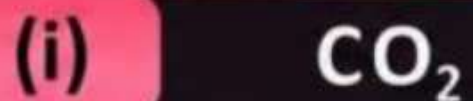
$$= 6 - 7$$

$$= -1$$

Example



Write Lewis structures or electron dot symbols of the following molecules :





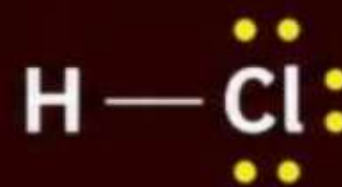
Limitations of Octet Rule



1 Compounds containing Hydrogen (H) :

- Hydrogen has only 1 electron in its valence shell.
- It needs only 1 more electron to achieve the configuration of Helium (duplet).
- Hence, hydrogen completes a duplet, not an octet.

e.g.



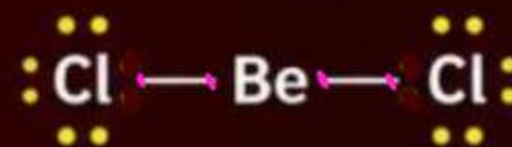
Note :

Hydrogen follows duplete rule (2 electrons).

2 Electron-deficient Compounds (BeCl_2 , BF_3 , AlCl_3 , etc.) :

- The central atom has less than 8 electrons.
- Such molecules are called hypovalent species or electron-deficient compounds.

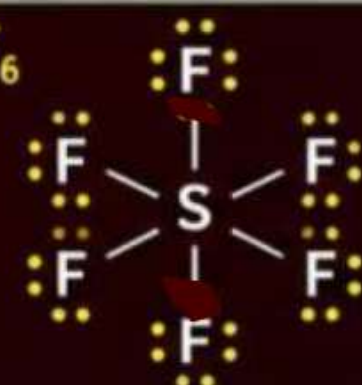
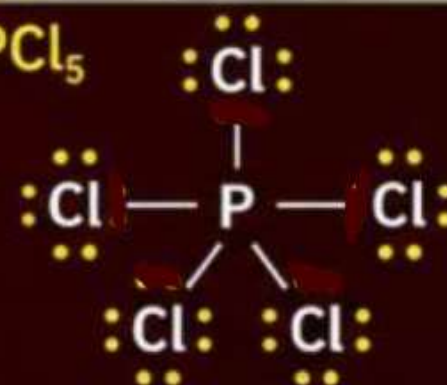
e.g.



3 Electron-rich Compounds (PCl_5 , SF_6 , IF_7 , etc.) :

- The central atom has more than 8 electrons.
- These are called hypervalent species or electron-rich compounds.

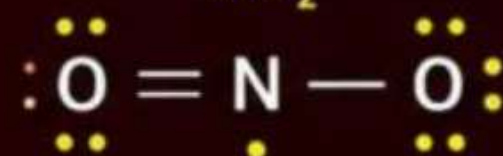
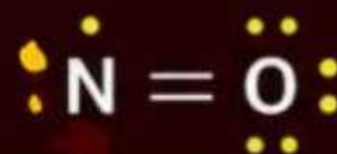
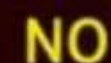
e.g.



4 Odd-electron Molecules (NO , NO_2) :

- Molecules having an odd number of electrons cannot satisfy the octet rule for all atoms.

e.g.





Resonance

(Imp).

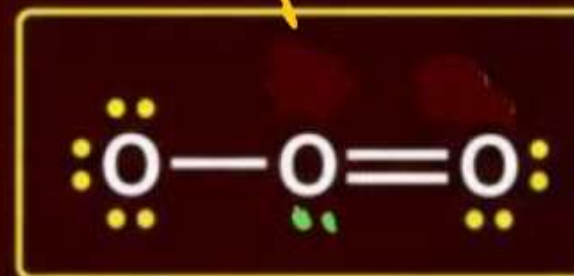
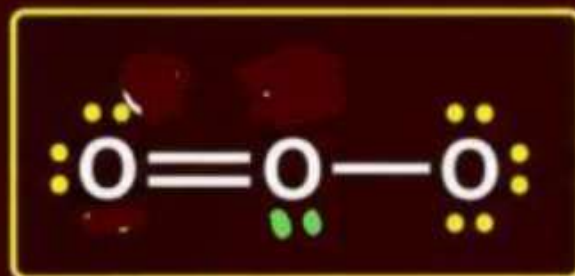


Key Points

- 1 When one **Lewis structure cannot explain** a molecule's properties, more than one structure is drawn.
- 2 These are called **canonical** or **resonating forms**.
- 3 The actual molecule is a **resonance hybrid**, which is a combination of all canonical forms.
- 4 This phenomenon is called **resonance**.

Example : O₃ (Ozone)

- O₃ (ozone) – bond lengths are equal due to resonance.



Resonance hybrid



Both O—O bonds are equal

Important Notes

- 1 Resonating structures are **imaginary**; only the hybrid exists.
- 2 **Resonance energy** = Energy of the most stable canonical form – Energy of resonance hybrid.

Note :

Resonance hybrid is more stable than any single canonical form.



Rules for Resonance



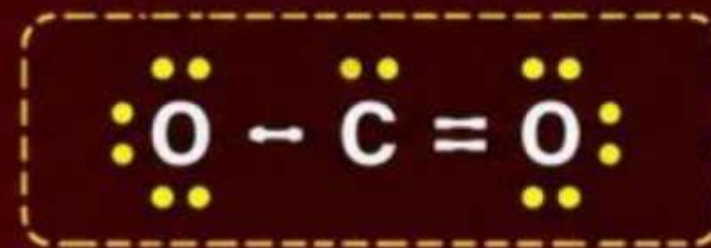
1

Atoms' positions remain the same.



2

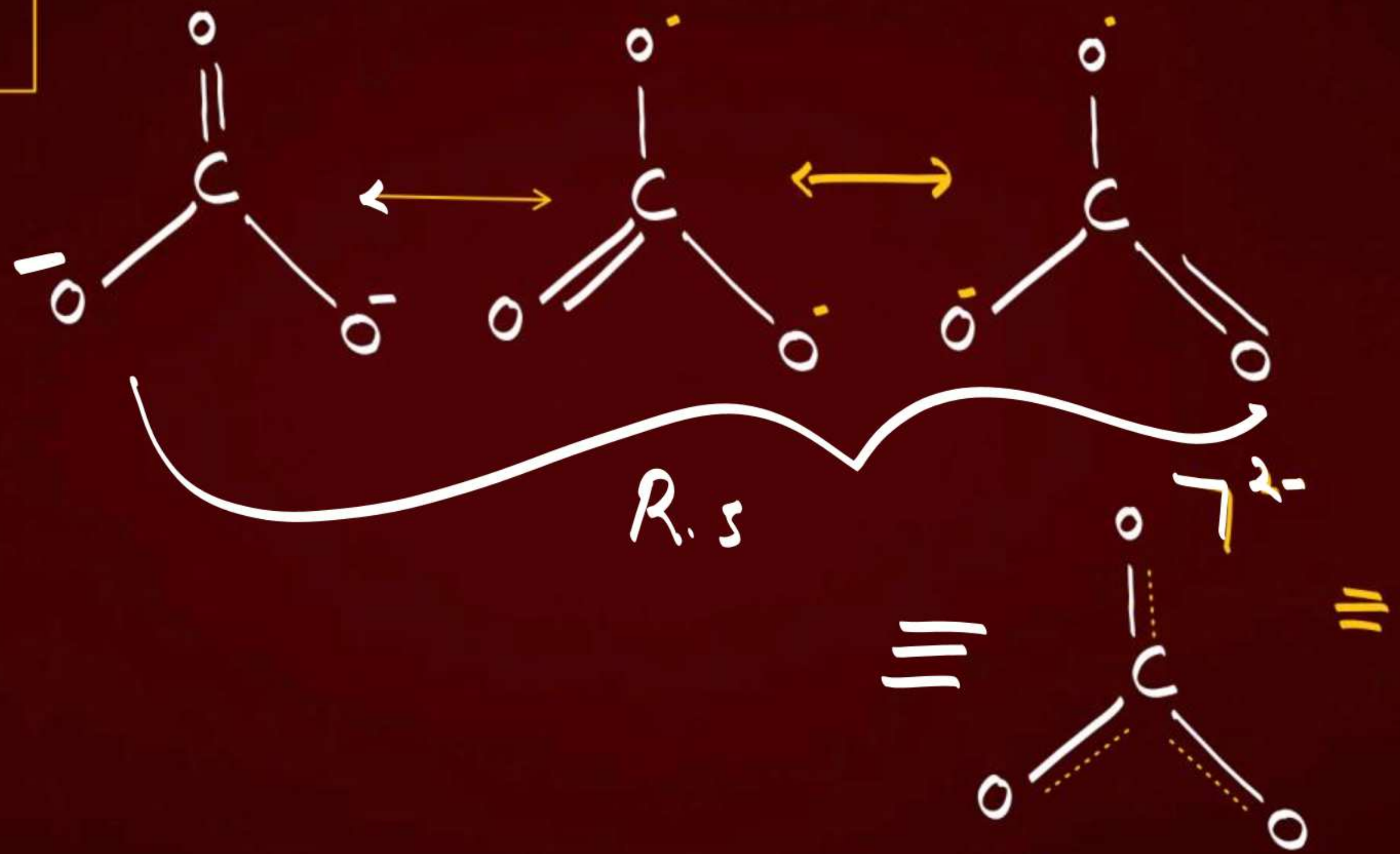
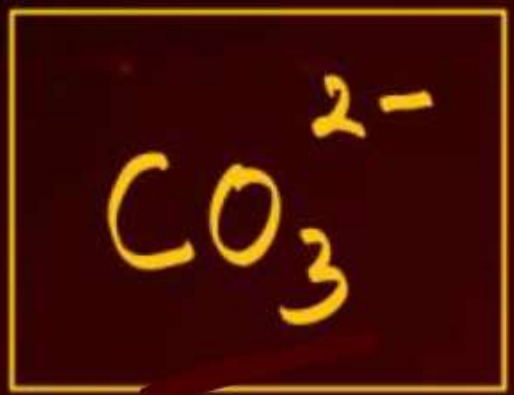
The **total number of electrons** (bonding + lone pairs) remains the same.



3

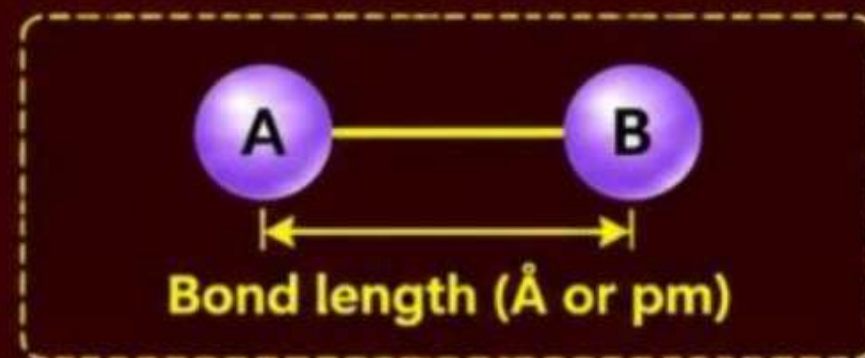
The **energies** of all contributing structures should be nearly equal.







- Distance between nuclei of two bonded atoms.
- **Unit:** Å (angstrom) or pm (picometre).



Bond length is the equilibrium **distance** between the nuclei of two bonded atoms.

(a) Size of atoms

- Larger atoms → longer bond length.
- Smaller atoms → shorter bond length.

gth

512

gth.

F
C
B
I

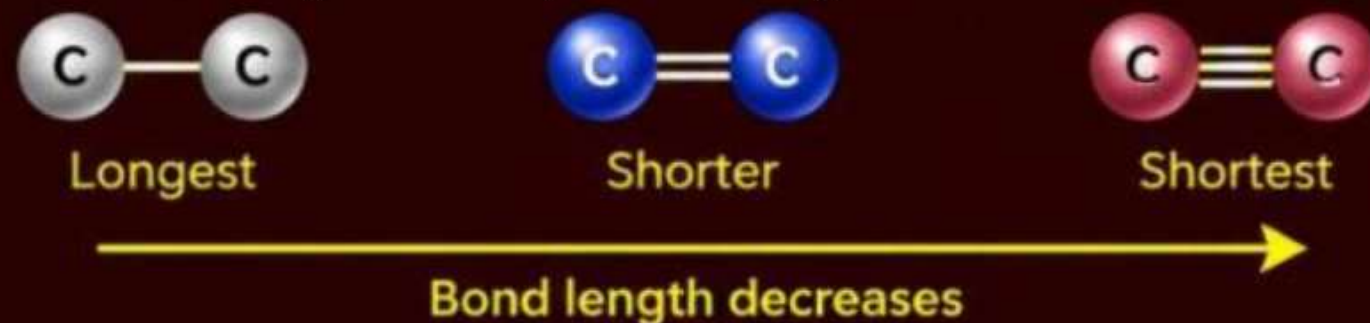
(b) Bond multiplicity

- More bonds → shorter bond length.
- Bond length decreases in the order:

Example : $\text{H}-\text{F} < \text{H}-\text{Cl} < \text{H}-\text{Br} < \text{H}-\text{I}$



Example : $C - C > C = C > C \equiv C$





Bond Parameters

$$\text{Bond Enthalpy} = \frac{BE_1 + BE_2}{2} =$$



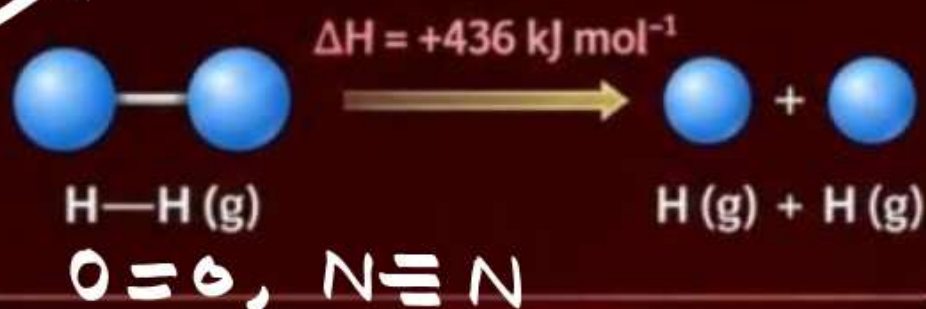
2. Bond Enthalpy

The amount of **energy required to break** one mole of a particular bond in the **gaseous state** into ~~atoms~~.

Unit: **kJ mol⁻¹**

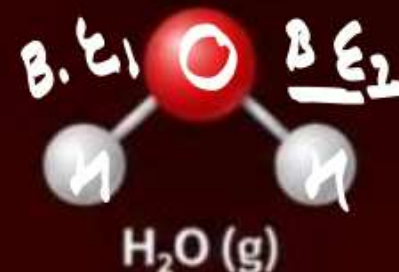
Diatomic molecules

Bond enthalpy = **Bond dissociation energy.**



Polyatomic molecules

Bond enthalpy = **Average of similar bond energies.**

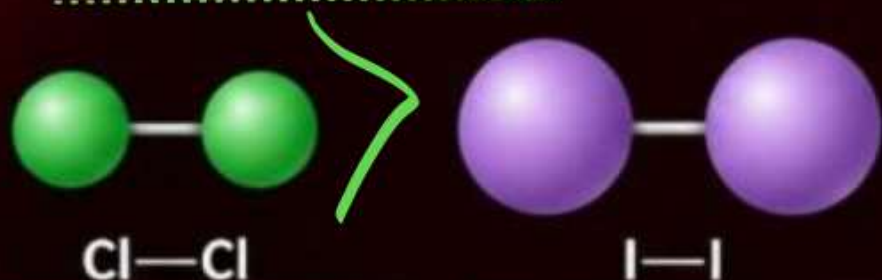


Both O-H bonds have the same bond enthalpy, so we take their average.

Factors affecting bond enthalpy

(a) Size of atoms

Larger atoms → **weaker** bond → **less** bond enthalpy.

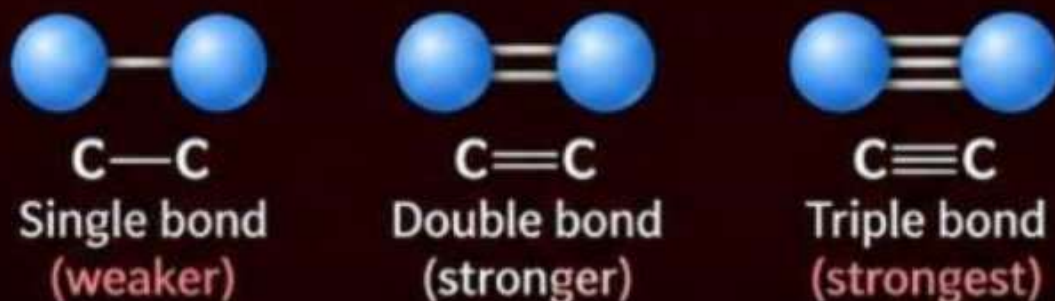


Stronger bond
(higher enthalpy)

Weaker bond
(lower enthalpy)

(b) Bond multiplicity

More number of bonds → **stronger** bond → **higher enthalpy.**



Bond enthalpy increases

(c) Lone pairs

More lone pairs → **more repulsion** → **lower** bond enthalpy.



Lone pairs create repulsion, which weakens the bond.



Bond Parameters

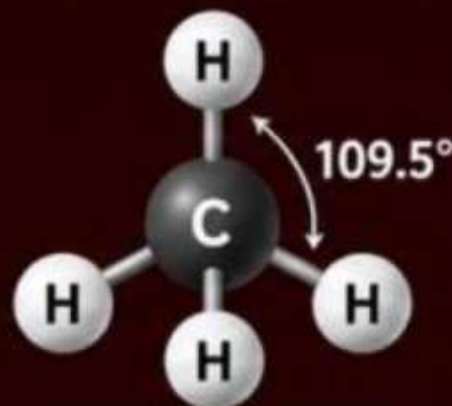


3. Bond Angle

- Angle between **two bonded pairs** of electrons.
- Expressed in **degrees (°)**.
- As **lone pairs increase** → bond angle decreases (due to more repulsion).

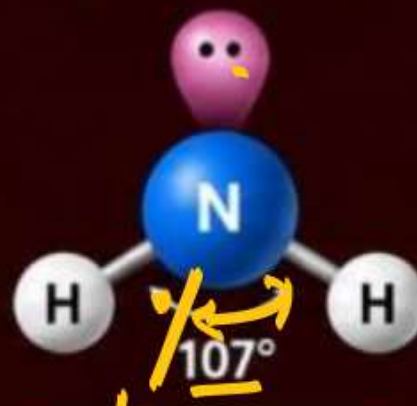
Examples

CH4
(109.5°)



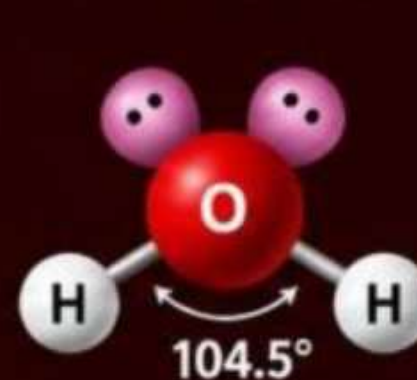
Tetrahedral

NH3
(107°)



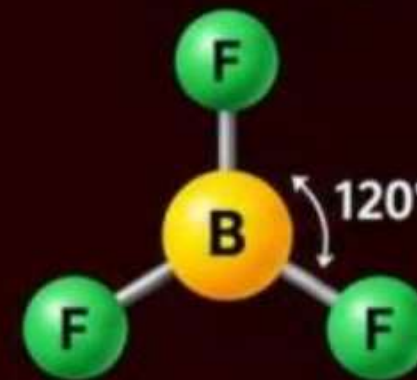
Trigonal pyramidal

H2O
(104.5°)



Bent (V-shaped)

BF3
(120°)



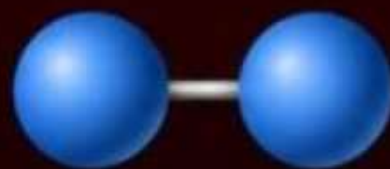
Trigonal planar

4. Bond Order

- Number of chemical bonds between two atoms.
- Higher bond order means **stronger bond** (higher bond energy) and **shorter bond length**.

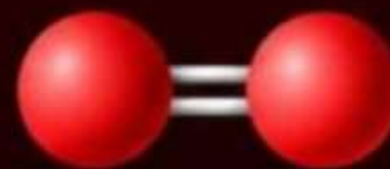
Examples

H2 (Bond order = 1)



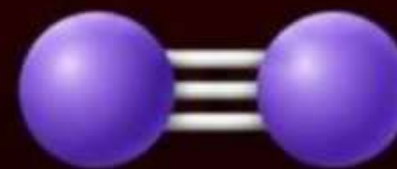
Single bond

O2 (Bond order = 2)



Double bond

N2 (Bond order = 3)



Triple bond

Relation

Bond order ↑



Bond energy ↑

More bonds → stronger bond
→ **higher bond energy**.



Dipole Moment (μ)

What is Dipole Moment?

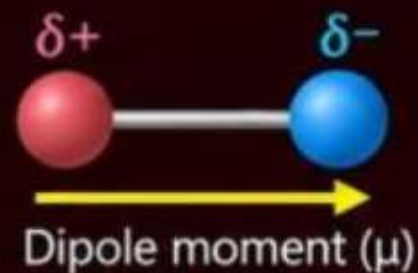
It measures the **polarity of a molecule**, i.e. how much positive and negative charges are separated.

Formula

$$\mu = q \times d$$

q = charge (C)

d = distance between charges (m)



Unit

Debye (D)

$$1 \text{ D} = 3.33 \times 10^{-30} \text{ C}\cdot\text{m}$$

Nature

- **Vector** quantity.
- Direction is from **positive to negative** charge.

General Trends

1 Diatomic molecules

Dipole moment **increases** with increasing **electronegativity difference** (ΔEN).

Example:



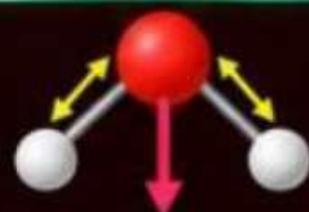
Order: $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$

2 Polyatomic molecules

Dipole moment depends on **individual bond dipoles** and the **molecular geometry** (vector sum).

Examples:

H_2O (Bent)



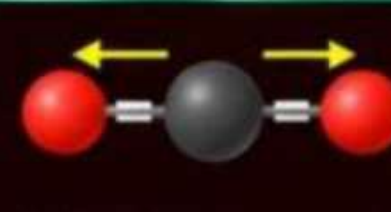
Net dipole moment
(**non-zero**)

NH_3 (Trigonal pyramidal)



Net dipole moment
(**non-zero**)

CO_2 (Linear)



Net dipole moment
(**zero**)

Same bonds can have zero or non-zero dipole moment depending on **geometry**.

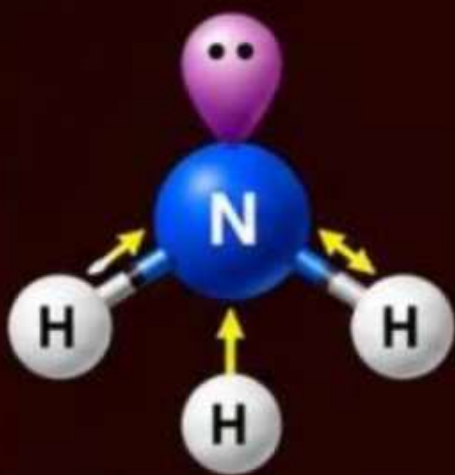


Dipole Moment (μ)



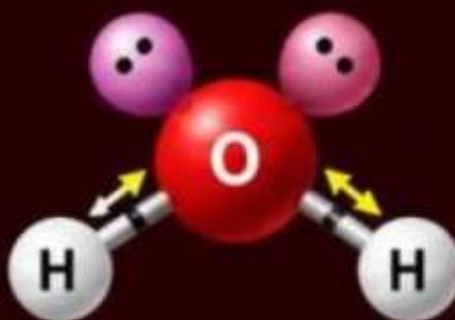
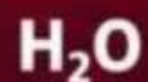
Examples

Dipole moment depends on **molecular geometry** (vector sum of bond dipoles)



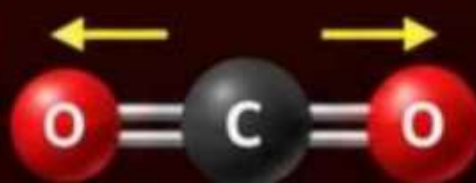
Trigonal pyramidal

$\mu \neq 0$
(Polar molecule)



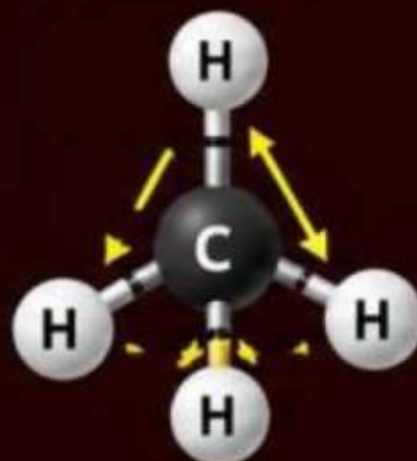
Bent (V-shaped)

$\mu \neq 0$
(Polar molecule)



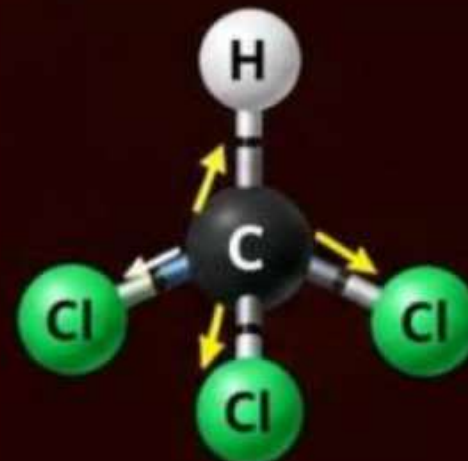
Linear

$\mu = 0$
(Non-polar molecule)



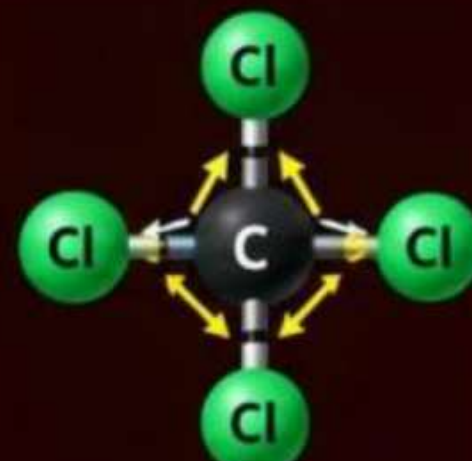
Tetrahedral
(symmetric)

$\mu = 0$
(Non-polar molecule)



Tetrahedral
(asymmetric)

$\mu \neq 0$
(Polar molecule)



Tetrahedral
(symmetric)

$\mu = 0$
(Non-polar molecule)

Same type of bonds can give **zero** or **non-zero** dipole moment depending on the **geometry** of the molecule.

Dipole moment Ka Devi Nuksa.

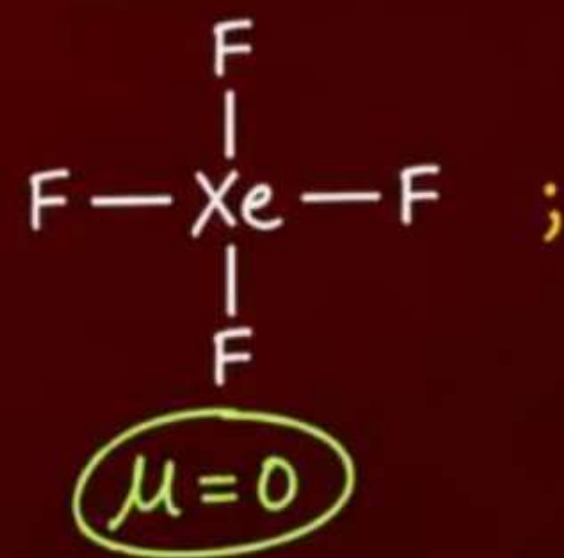
If side atom same
 & no lone pair on CA.
 then $\mu = 0$ (non polar)

e.g. PCl_5 , PF_5 , IF_7 , SF_6 , CH_4 etc.

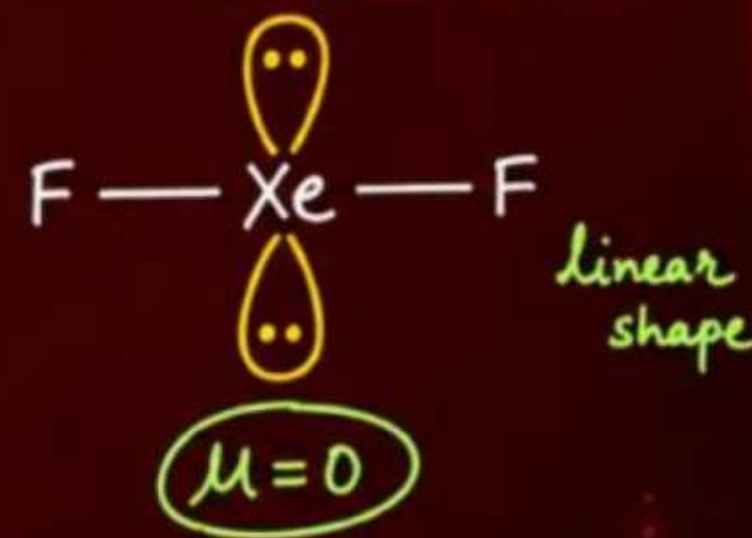
If side atom same & lp present.

Only in square planar & Linear shape.
 then $\mu = 0 \Rightarrow$ non-polar molecule.

(i) Square planar



(ii) Linear shape

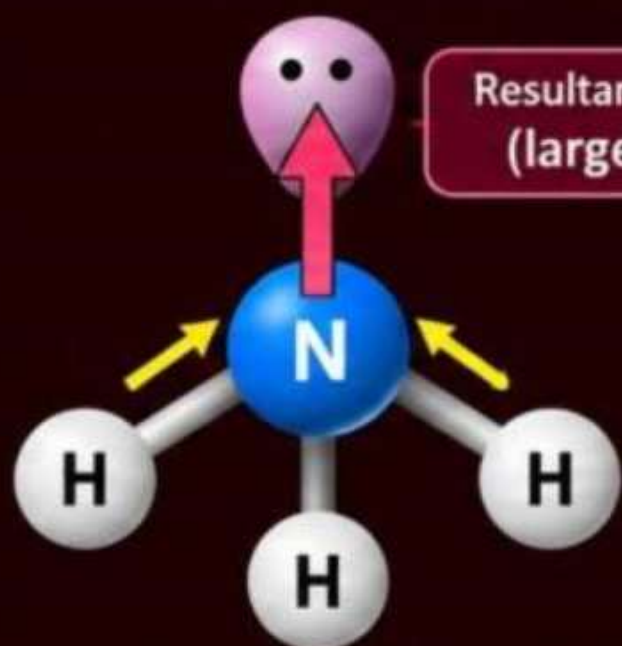




Dipole Moment (μ)



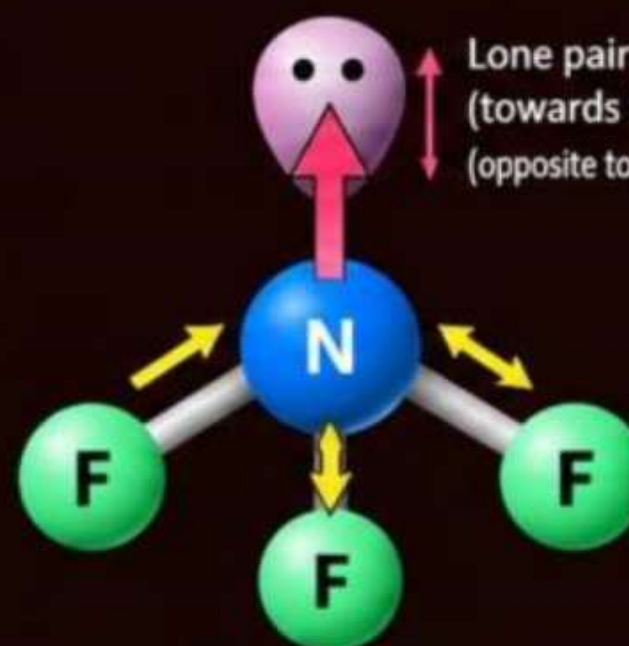
Comparison of NH_3 and NF_3



Resultant μ
(large)

- **Bond dipoles** ($\text{N} \leftarrow \text{H}$) and lone pair dipole act in the **same direction** (towards N).
- Hence, the dipole moment is **large**.

Trigonal pyramidal



Lone pair dipole
(towards N)
(opposite to bond dipoles)

- Bond dipoles ($\text{N} \rightarrow \text{F}$) **oppose** the lone pair dipole.
- Hence, the dipole moment is **smaller** due to partial cancellation.

Trigonal pyramidal

$$\therefore \mu(\text{NH}_3) > \mu(\text{NF}_3)$$

Q. Which of the following obeys octet rule.





Valence Shell Electron Pair Repulsion (VSEPR) Theory



6). Order of Repulsion $\rightarrow lp-lp > bp-lp > bp-bp$

Introduction

- **Idea given by:** Sidgwick and Powell in 1940.
- **Improved by:** Nyholm and Gillespie in 1957.

Basic Concept

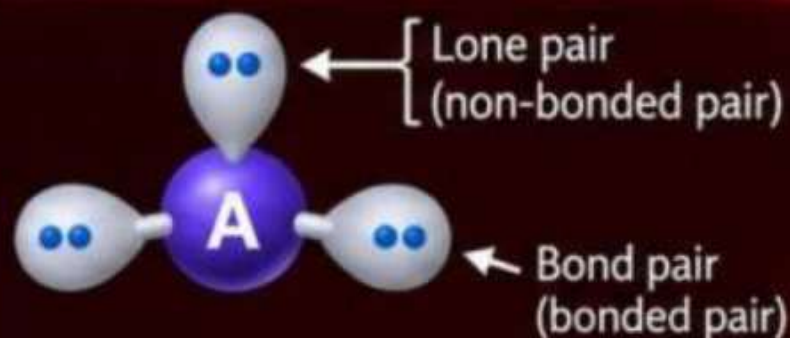
The shape of a molecule depends on how **electron pairs (bonded or lone pairs)** arrange themselves around the **central atom**.

Main Points of VSEPR Theory

- 1 The shape of a molecule depends on the **number of electron pairs** (bond pairs + lone pairs) around the **central atom**.
- 2 Since electron clouds are negatively charged, they **repel each other**.
- 3 To reduce repulsion, electron pairs try to stay **as far apart as possible**.
- 4 We imagine the **valence shell as a sphere**, and the electron pairs spread out on its surface at **maximum distance**.
- 5 A double or triple bond is treated like one single pair (called a **super pair**).

Key Ideas with Visuals

1 Electron pairs around central atom



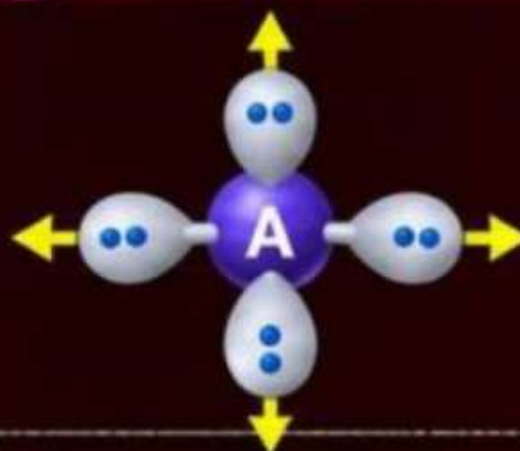
Shape depends on the **number of electron pairs** around the **central atom**.

2 Repulsion between electron pairs



Since electron clouds are negatively charged, they **repel each other**.

3 Stay as far apart as possible



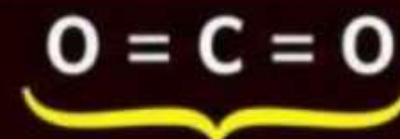
To reduce repulsion, electron pairs try to stay **as far apart as possible**.

4 Valence shell as a sphere



We imagine the **valence shell as a sphere**, and the electron pairs spread out on its surface at **maximum distance**.

5 Double or triple bond as one pair



Two double bonds = two electron pairs

Treated as one pair (**super pair**).



PREDICTING THE SHAPES OF MOLECULES

1 Calculation of TEP (Total electron pair)

$$\text{TEP} = \frac{1}{2} (\text{V} + \text{M} - \text{C} + \text{A})$$



V = Valence electrons of central atom.



M = Number of monovalent side atoms (except nitrogen & oxygen family).



C = Charge on cation.



A = Charge on anion.

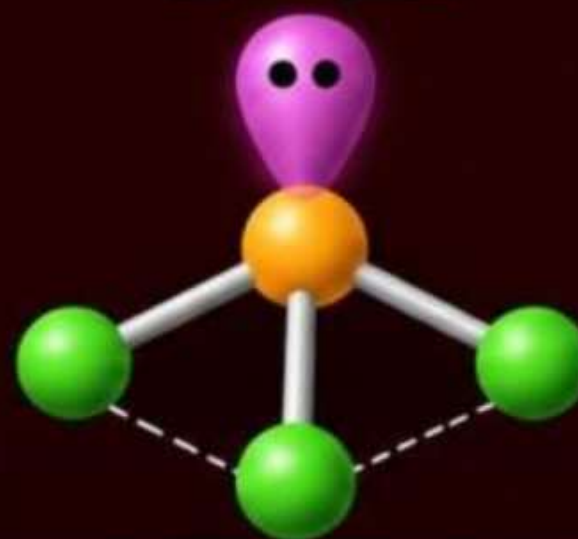


lp = Lone pair (non-bonding pair)

bp = Bonding pair

GEOMETRY

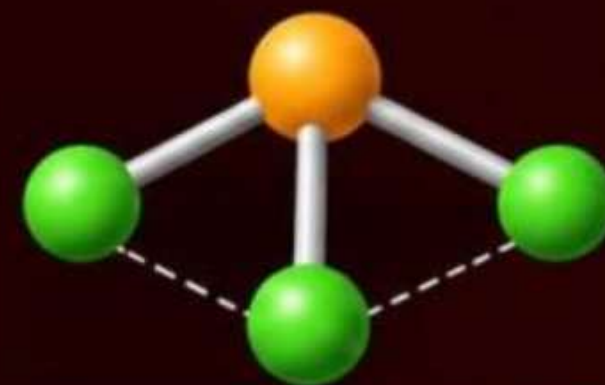
We include both **lp** & **bp**.



Example:
Trigonal
Pyramidal

SHAPE

We only include **bp**.



Example:
Pyramidal

TEP

2

3

4

5

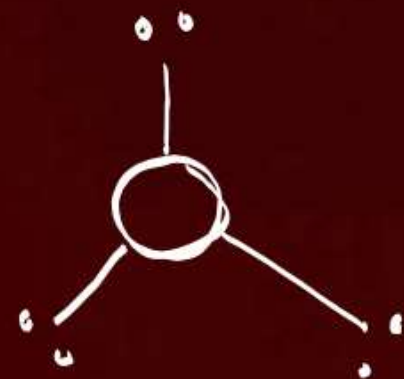
6.

Geometry

Linear



Trigonal Planar



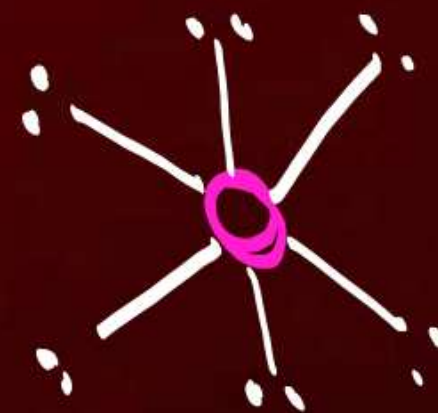
Tetrahedral



Trigonal bi pyramidal



Octahedral



PREDICTING THE SHAPES OF MOLECULES

1 $TEP = 2 (lp + bp)$

Geometry = **Linear** $:\text{—O—}:$

$lp = 0, bp = 2$

Shape $\rightarrow$ **Linear**

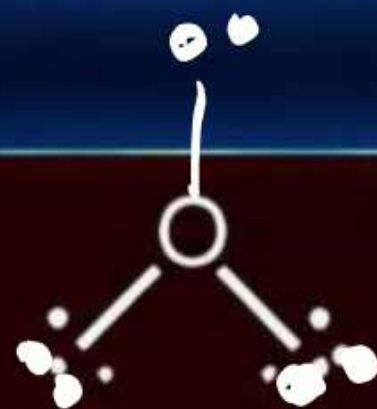
e.g. F—Be—F



LINEAR

2 $TEP = 3 (lp + bp = 3)$

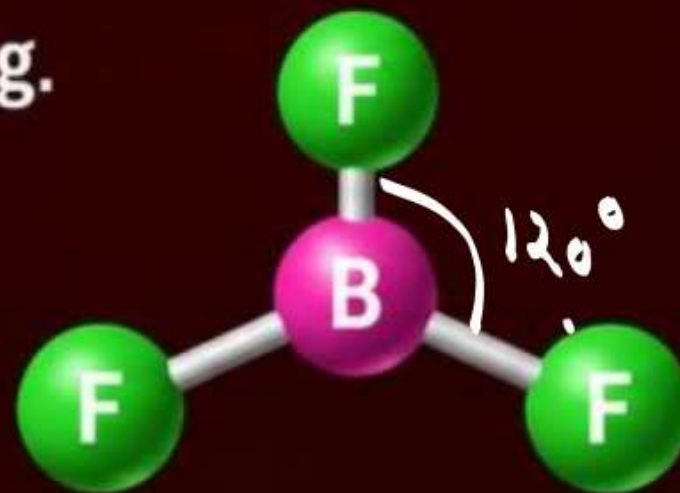
Geometry = **Trigonal planar**



$lp = 0, bp = 3$

Shape $\rightarrow$ **Trigonal planar**

e.g.

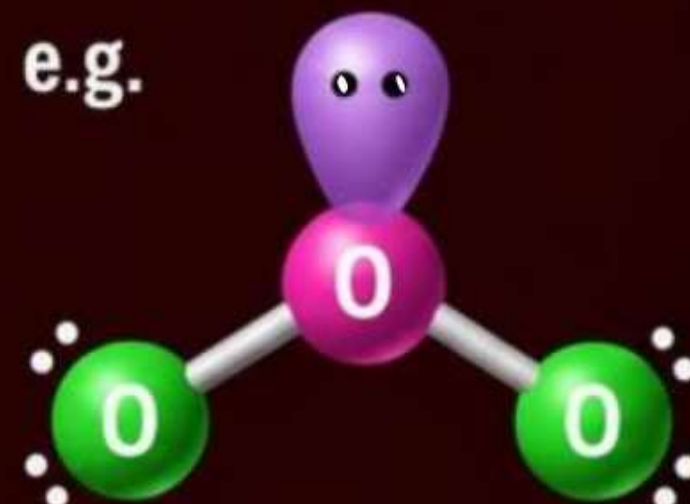


TRIGONAL PLANAR

$lp = 1, bp = 2$

Shape $\rightarrow$ **Bent**

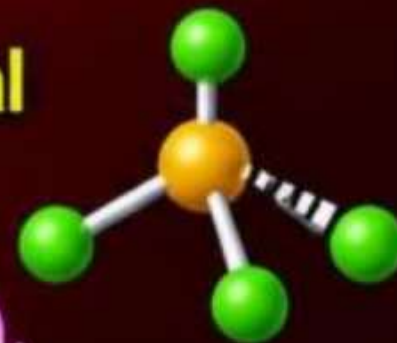
e.g.



BENT

3) TEP = 4 (lp + bp = 4)

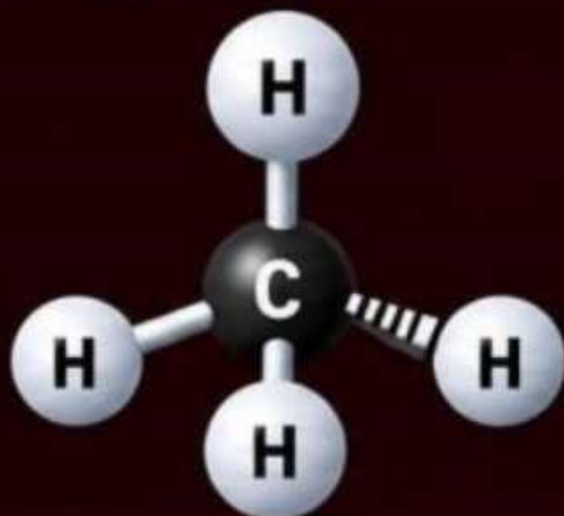
Geometry → **Tetrahedral**



lp = 0, bp = 4

Shape → **Tetrahedral**

e.g. **CH₄**

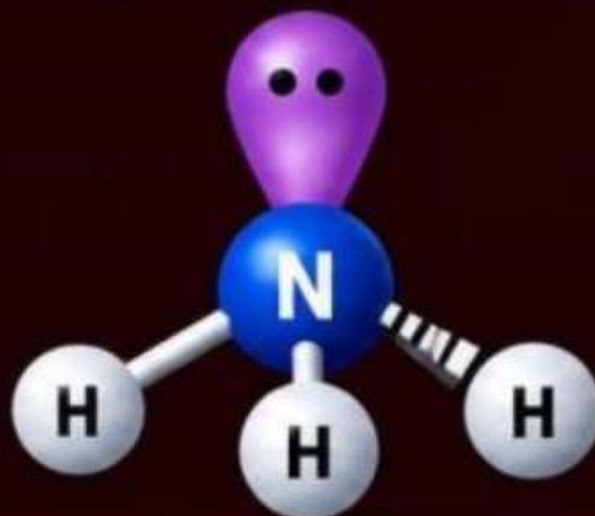


TETRAHEDRAL

lp = 1, bp = 3

Shape → **Pyramidal shape**

e.g. **NH₃**

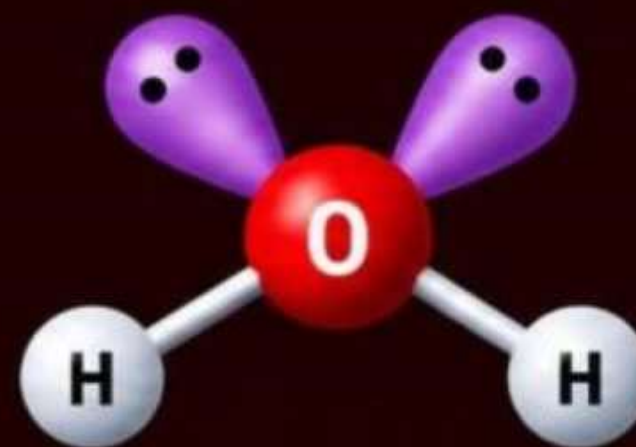


PYRAMIDAL

lp = 2, bp = 2

Shape → **Bent or V-shape**

e.g. **H₂O**



BENT or V-SHAPE

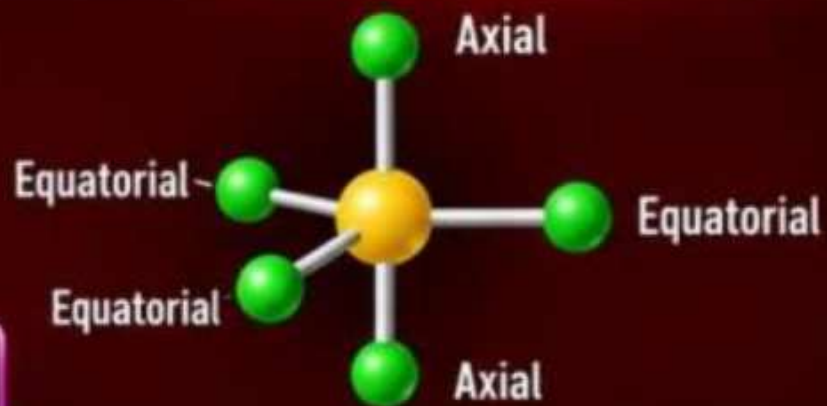
lp = Lone pair
(Non-bonding pair)

bp = Bonding pair

Predicting the Shapes of Molecules

4) TEP = 5 (lp + bp)

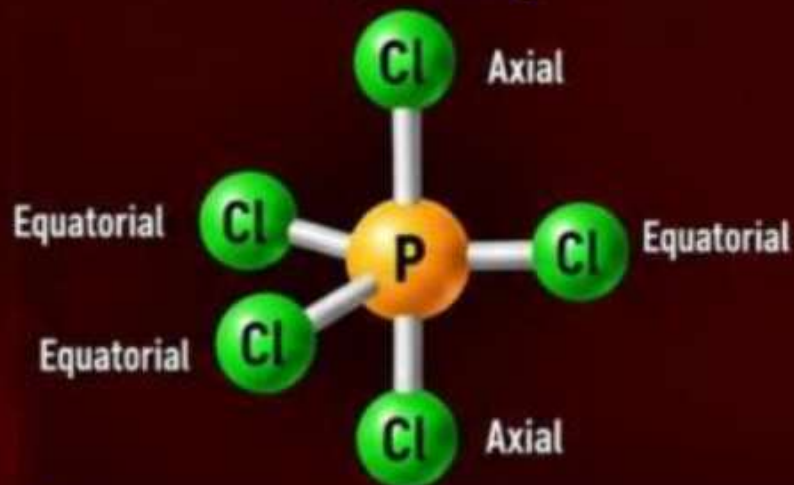
Geometry = Trigonal bipyramidal (TBP)



0 lp + 5 bp

Shape → Trigonal bipyramidal (TBP)

e.g. PCl_5

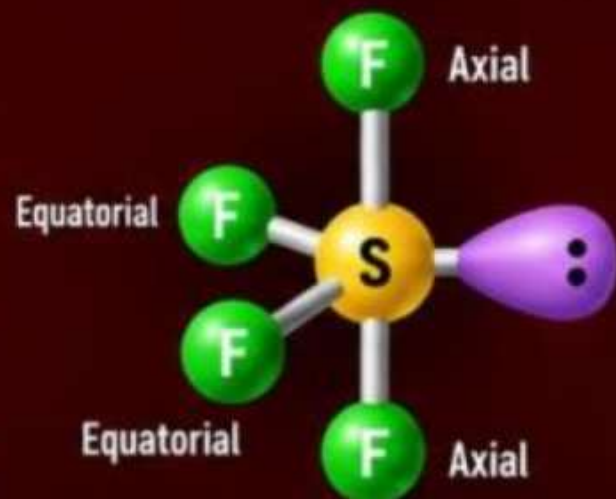


TRIGONAL BIPYRAMIDAL

1 lp + 4 bp

Shape → See-saw

e.g. SF_4

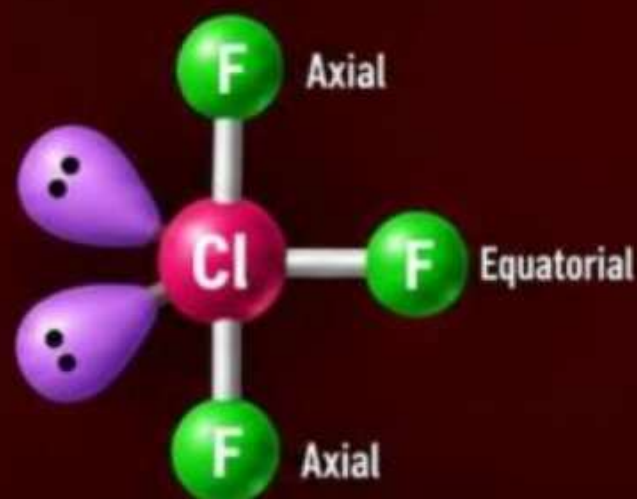


SEE-SAW

2 lp + 3 bp

Shape → T-shape

e.g. ClF_3



T-SHAPE

3 lp + 2 bp

Shape → Linear

e.g. XeF_2



LINEAR

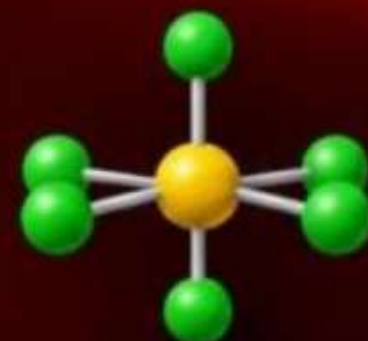
lp = Lone pair (Non-bonding pair)

bp = Bonding pair

 = Lone pair

$$\text{TEP} = 6 (\text{lp} + \text{bp})$$

Geometry = **Octahedral**



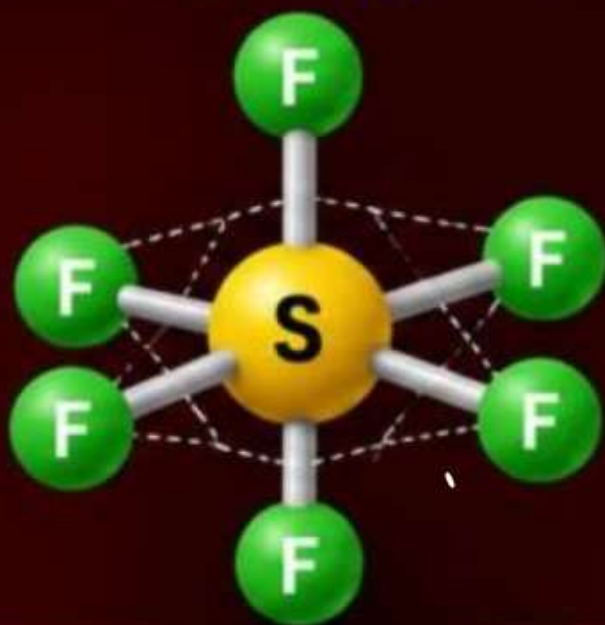
Ideal octahedral
arrangement
(90°)



0 lp + 6 bp

Shape → **Octahedral**

e.g. **SF₆**

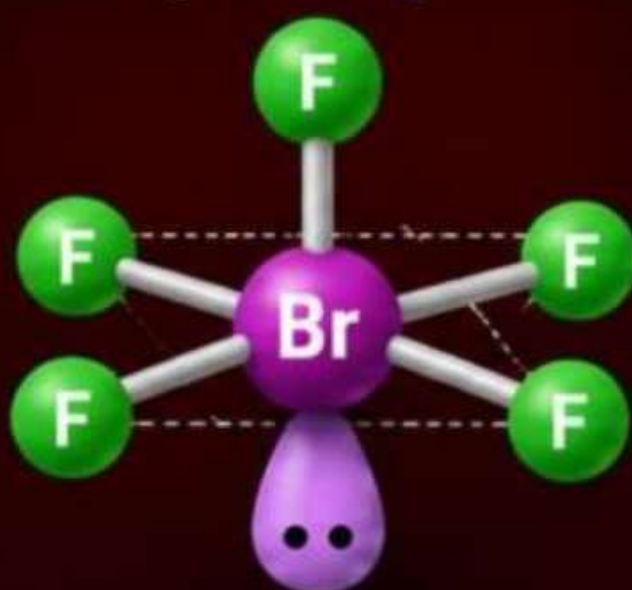


OCTAHEDRAL

1 lp + 5 bp

Shape → **Square pyramidal**

e.g. **BrF₅**

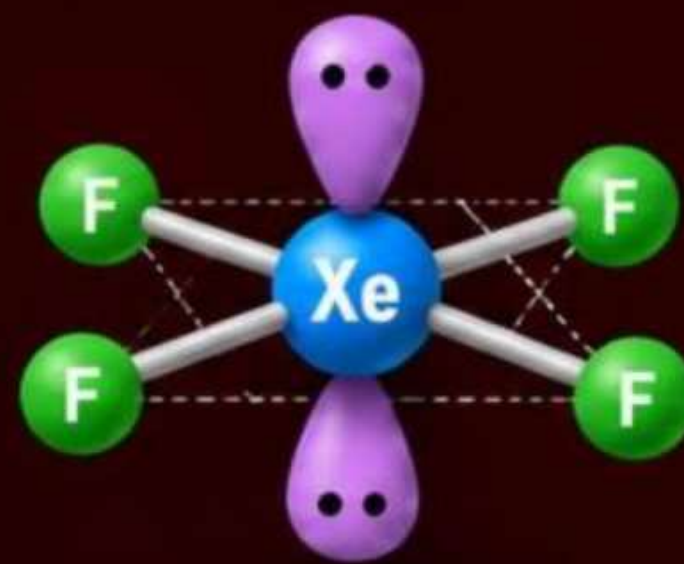


SQUARE PYRAMIDAL

2 lp + 4 bp

Shape → **Square planar**

e.g. **XeF₄**



SQUARE PLANAR

lp = Lone pair (Non-bonding pair)

bp = Bonding pair



= Lone pair

TEP

Geometry

SHAPE.

2 → linear → linear.

3 → TRIGONAL PLANAR → Trigonal planar & bent.

4 → Tetrahedral → Tetrahedral, Pyramidal, Bent.

5 → TBP → TBP, See-saw, T-shape, linear.

6 → Octahedral. → Octahedral, Square pyramidal, Square planar.

1. $\text{SF}_4 \rightarrow$ SF_6
2. $\text{XeF}_4 \rightarrow$
3. $\text{CO}_3^{2-} \rightarrow$ BrF_5
4. ~~scribbled out~~ $\rightarrow$
5. $\text{SO}_3 \rightarrow$
6. $\text{ClF}_3 \rightarrow$
7. $\text{NH}_4^+ \rightarrow$
8. $\text{BCl}_3 \rightarrow$
9. $\text{PCl}_5 \rightarrow$
10. $\text{H}_2\text{O} \rightarrow$

Shape and Geometry $\Rightarrow$
 $\rightarrow \text{XeF}_4$

$$\text{TEP} = \frac{1}{2}(V + M - C + A)$$

$$= \frac{1}{2}(8 + 4) = \frac{12}{2} = 6$$

Geometry $\rightarrow$ Octahedral.

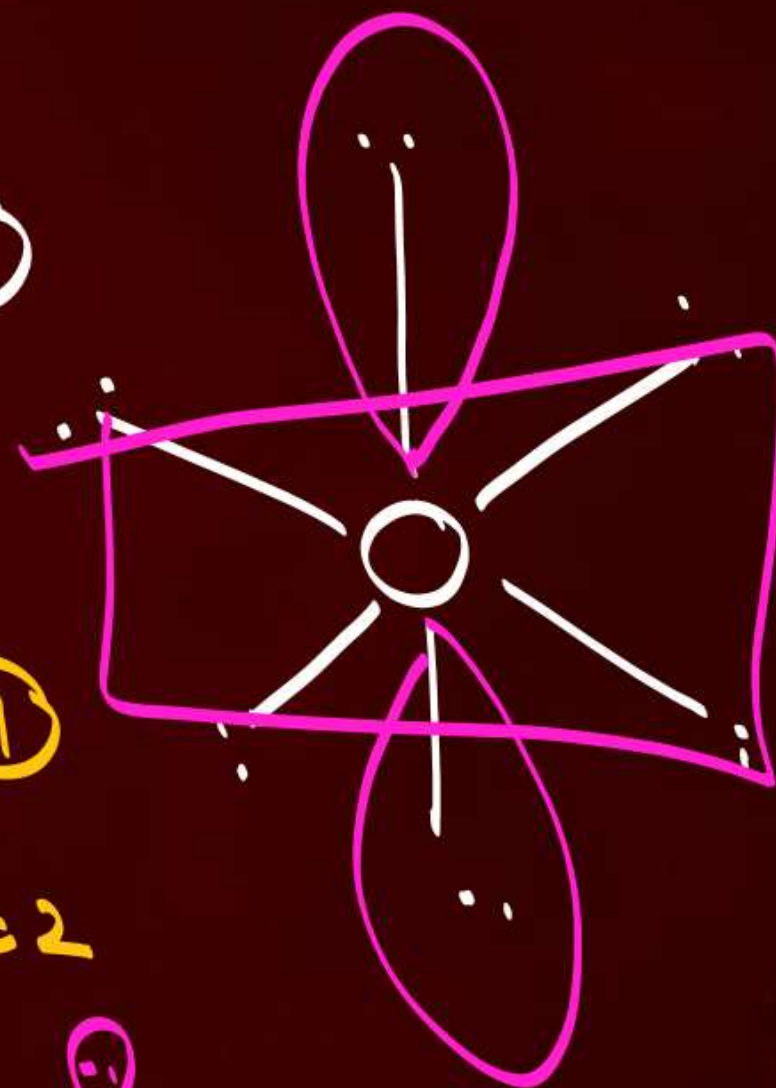
Shape $\Rightarrow$ (lp & bp).

$$\text{bp} = \text{No. of side atom} = 4$$

$$\text{lp} = \text{TEP} - \text{bp} = 6 - 4 = 2$$

$$\text{lp} = 2, \text{bp} = 4$$

Shape $\Rightarrow$ Square planar



b) CO_2

$$\text{TEP} = \frac{1}{2}(V + M - C + A)$$

$$= \frac{1}{2}(4 + 0) = \frac{4}{2} = 2$$

Linear

Shape $\Rightarrow$

$$b_p = \text{No. of side atom} = 2$$

$$l_p = \text{TEP} - b_p = 2 - 2 = 0$$

$$\underline{l_p = 0}, b_p = 2$$

Linear

Properties of **Ionic** and **Covalent** Compounds

Ionic Compounds



NaCl (Ionic lattice)

1 Physical State

- Solid (due to strong ionic bonds).



NaCl (solid)

2 Melting/Boiling Point

- Very high.



3 Solubility

- Easily soluble in water.



NaCl in water

4 Electrical Conductivity

- Solid: Non-conductor (ions can't move).
- Molten/solution: Good conductor (ions move freely).



Solid
(Not conducting)



Solution
(Conducting)

Covalent Compounds



H₂O (Covalent)

1 Physical State

- Can be solid, liquid, or gas (e.g., F₂, I₂, Br₂).



Solid
(I₂)



Liquid
(H₂O)



Gas
(F₂)

2 Melting/Boiling Point

- Low.



3 Solubility

- Polar ones dissolve in water.



Sugar (polar) in water

4 Electrical Conductivity

- Poor conductors of electricity.



Not conducting

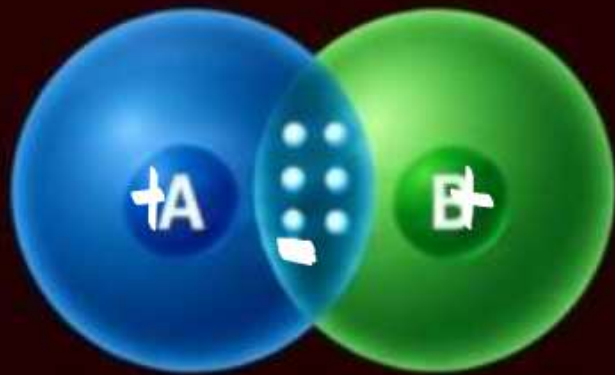


Valence Bond Theory

The valence bond theory was put forward by **Heitler and London** in **1927**. It was later improved and developed by **L. Pauling** and **J. C. Slater** in **1931**.

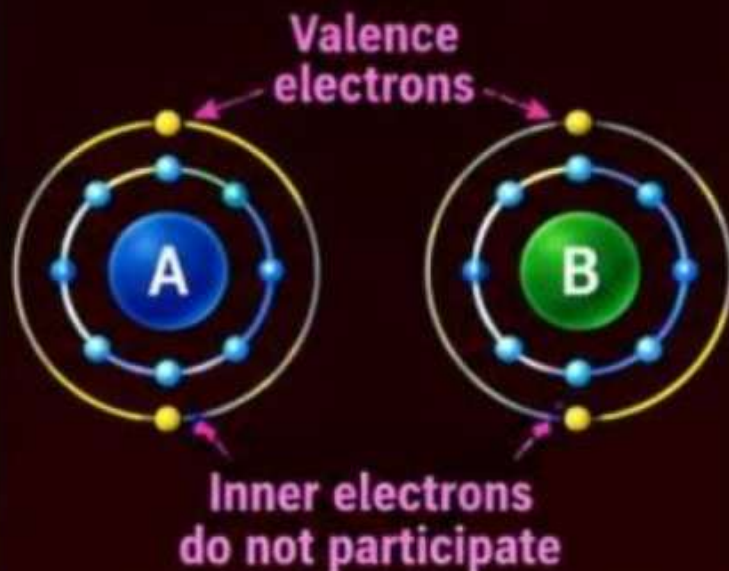
The basic assumptions of this theory are:

- 1** Atoms **do not lose their identity** even after the formation of the molecule.

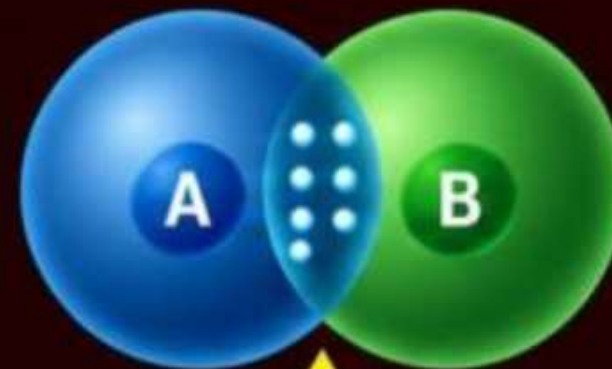


Atoms retain their identity.

- 2** The bond is formed due to the interaction of **only the valence electrons** as the two atoms come close to each other. The **inner electrons do not participate** in the bond formation.

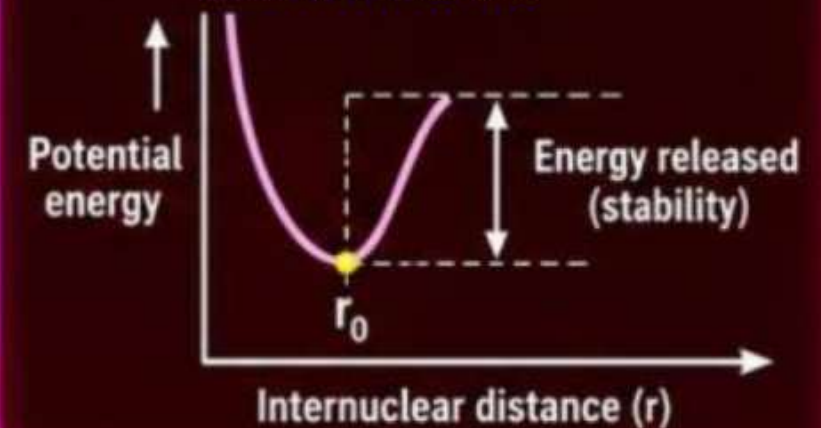


- 3** During the formation of bond, **only the valence electrons** from each bonded atom **lose their identity**. The other electrons **remain unaffected**.



Shared valence electrons lose their identity.

- 4** The stability of bond is accounted by the fact that the formation of bond is accompanied by **release of energy**. The molecule has **minimum energy** at a certain distance between the atoms known as **internuclear distance**. Larger the decrease in energy, stronger will be the bond formed.

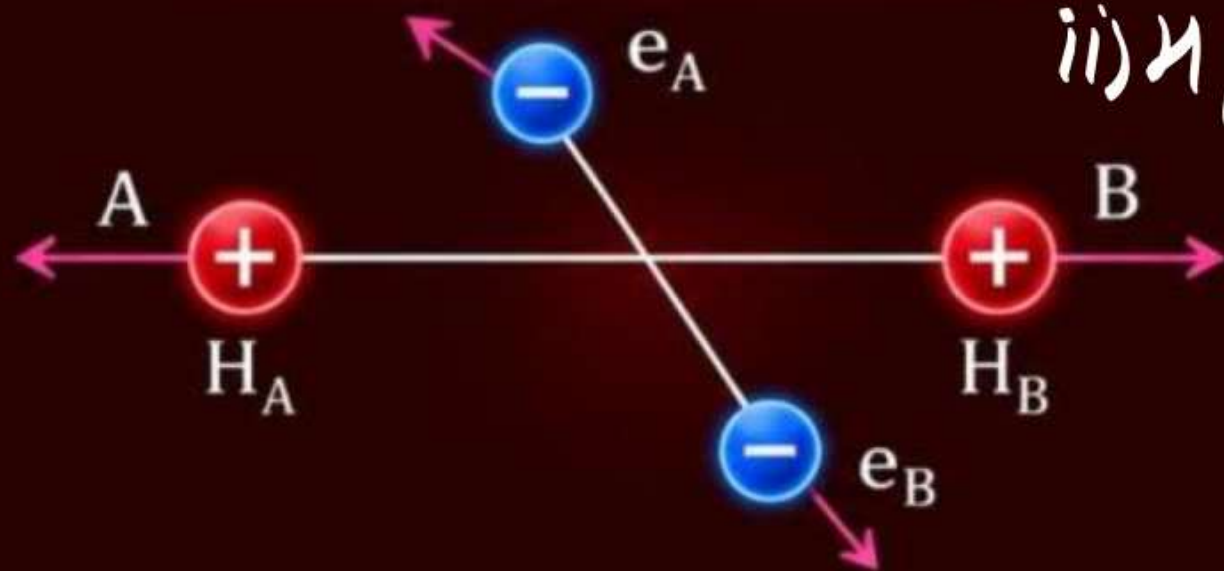




Valence Bond Treatment for H_2 Molecule

- Consider **two hydrogen atoms A and B** approaching each other having **nuclei H_A and H_B** and the corresponding **electrons e_A and e_B** respectively. When the two atoms come closer, attractive and repulsive forces begin to operate.

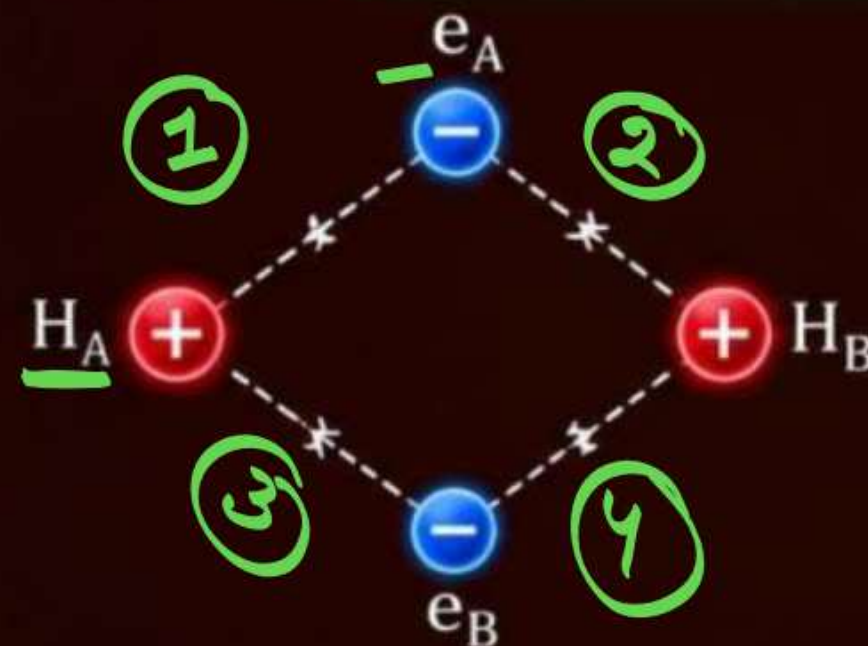
Repulsive Forces



- i) $e_A - e_B$
- ii) $H_A - H_B$

Repulsion exists between $H_A - H_B$, $e_A - e_B$ and each electron with the nucleus of the other atom.

Attractive Forces



- i) $H_A - e_A$
- 2) $H_B - e_A$
- 3) $H_A - e_B$
- 4) $H_B - e_B$

----- Old Forces
———— New Forces

Hence FOA dominate & H_2 formation takes place.

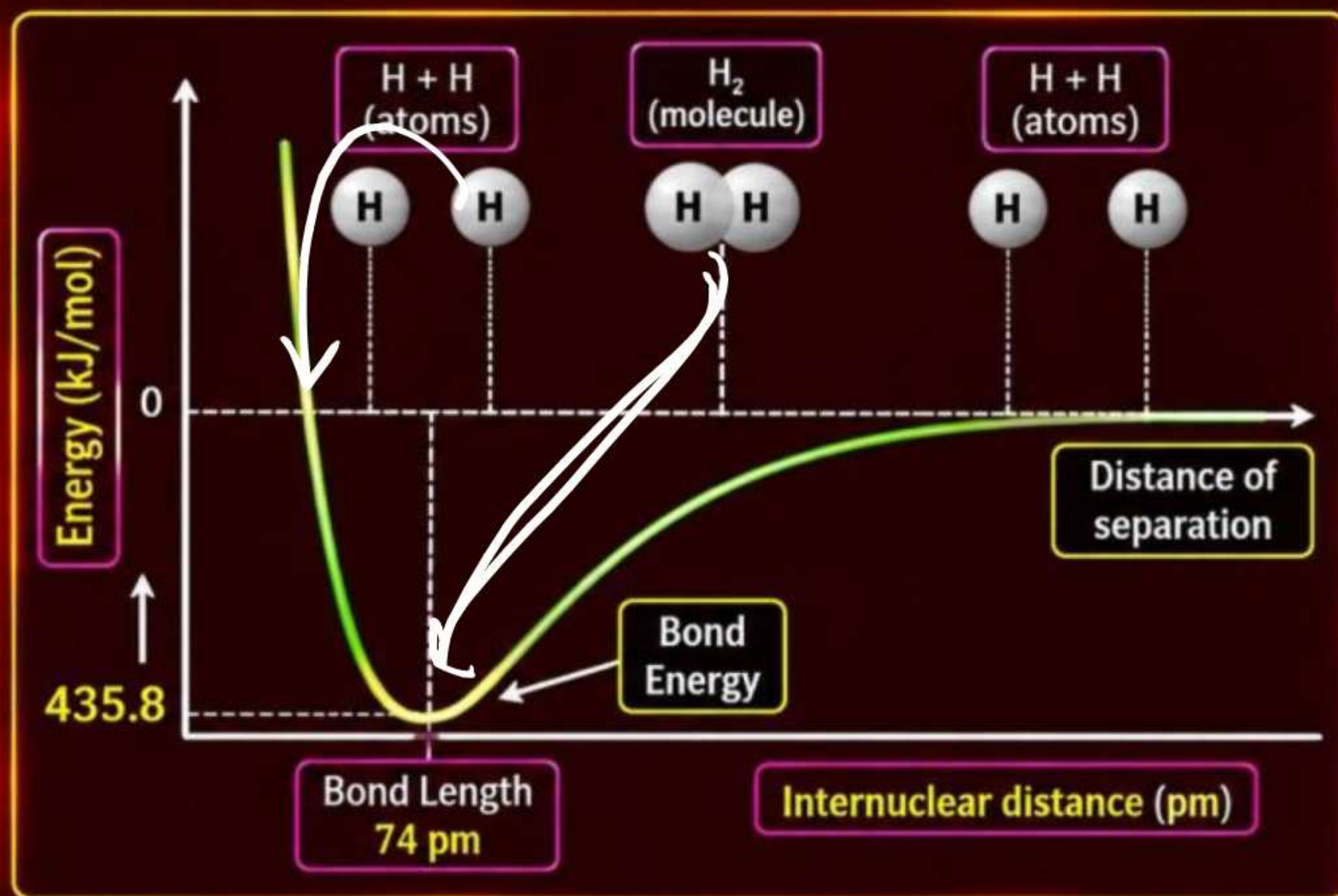
Attraction exists between each nucleus and the electron of the other atom ($H_A - e_B$ and $H_B - e_A$).



Valence Bond Treatment for H_2 Molecule

Potential Energy diagram

- When two hydrogen atoms bond to form a hydrogen molecule (H_2), **energy is released**, making the molecule **more stable** than separate atoms. This released energy is called **bond enthalpy** and corresponds to the **lowest point on an energy curve**. To break one mole of H_2 molecules back into atoms, **435.8 kJ** of energy is needed.



Key Takeaway: Formation of H_2 releases energy (bond enthalpy), and **435.8 kJ mol⁻¹** is required to break one mole of H_2 into atoms.

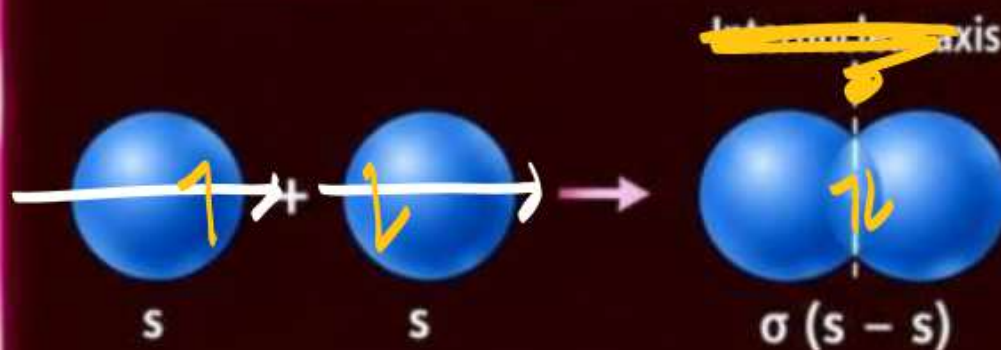


Types of Bond

Sigma (σ) bond is formed by **head-on (axial)** overlap of orbitals along the internuclear axis.

1. s - s Overlapping

Head-on overlap of two s-orbitals

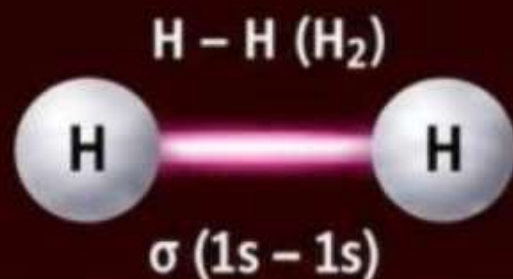


Example: H_2

Electronic configuration of H:

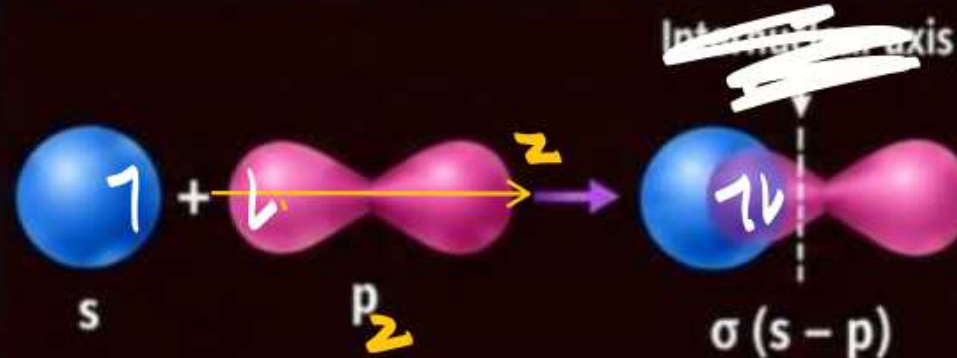
H ($Z = 1$) : $1s^1$

In H_2 : Each H atom contributes one 1s electron and overlap of two 1s orbitals forms a σ ($1s - 1s$) bond.



2. s - p Overlapping

Head-on overlap of s-orbital and p-orbital



Example: HF

Electronic configuration of H:

H ($Z = 1$) : $1s^1$

Electronic configuration of F:

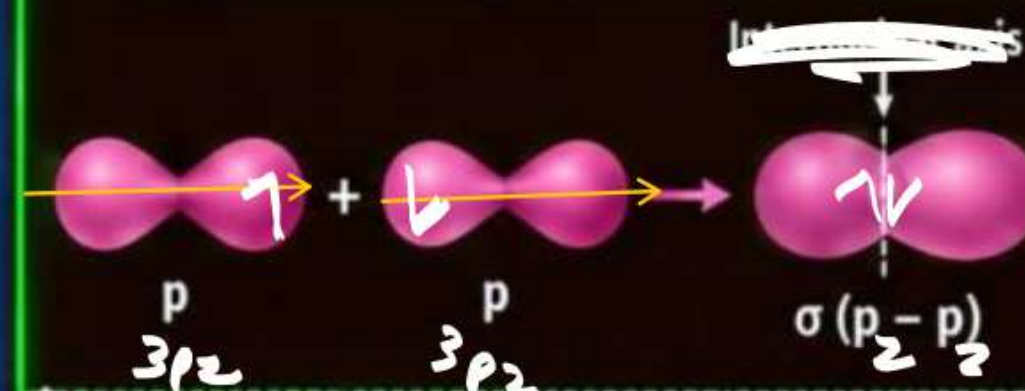
F ($Z = 9$) : $1s^2 2s^2 2p^5$ ($2p_x 2p_y 2p_z$)

In HF : 1s orbital of H overlaps with 2p orbital of F to form a σ ($1s - 2p$) bond.



3. p - p Overlapping

Head-on overlap of two p-orbitals

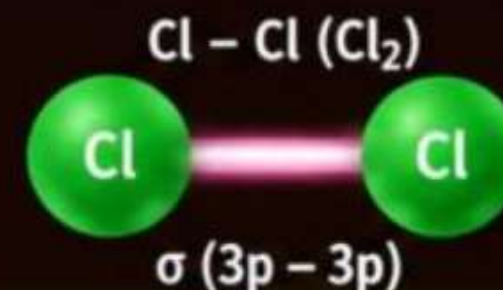


Example: Cl_2

Electronic configuration of Cl:

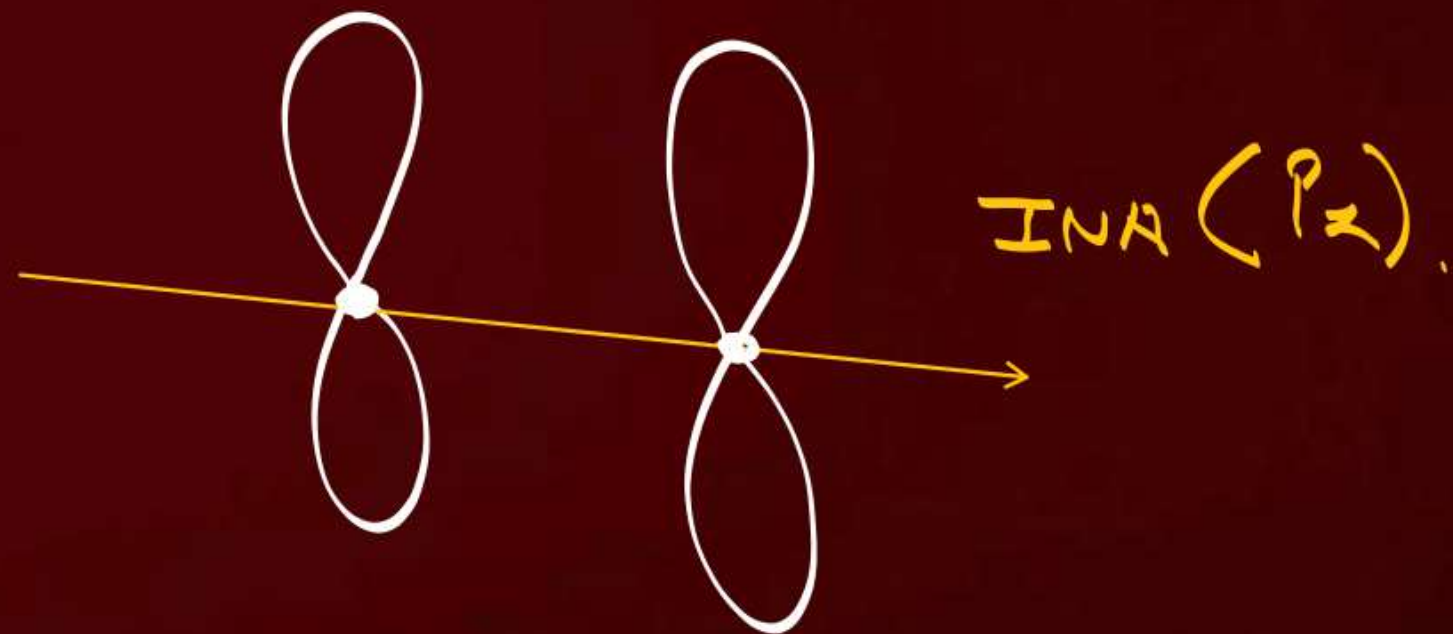
Cl ($Z = 17$) : $1s^2 2s^2 2p^6 3s^2 3p^5$ ($3p_x 3p_y 3p_z$)

In Cl_2 : One 3p orbital from each Cl atom overlaps to form a σ ($3p - 3p$) bond.



σ bond is stronger than π bond because of greater extent of overlap.

Internuclear axis $\rightarrow$ Axis passing through nucleus of two.





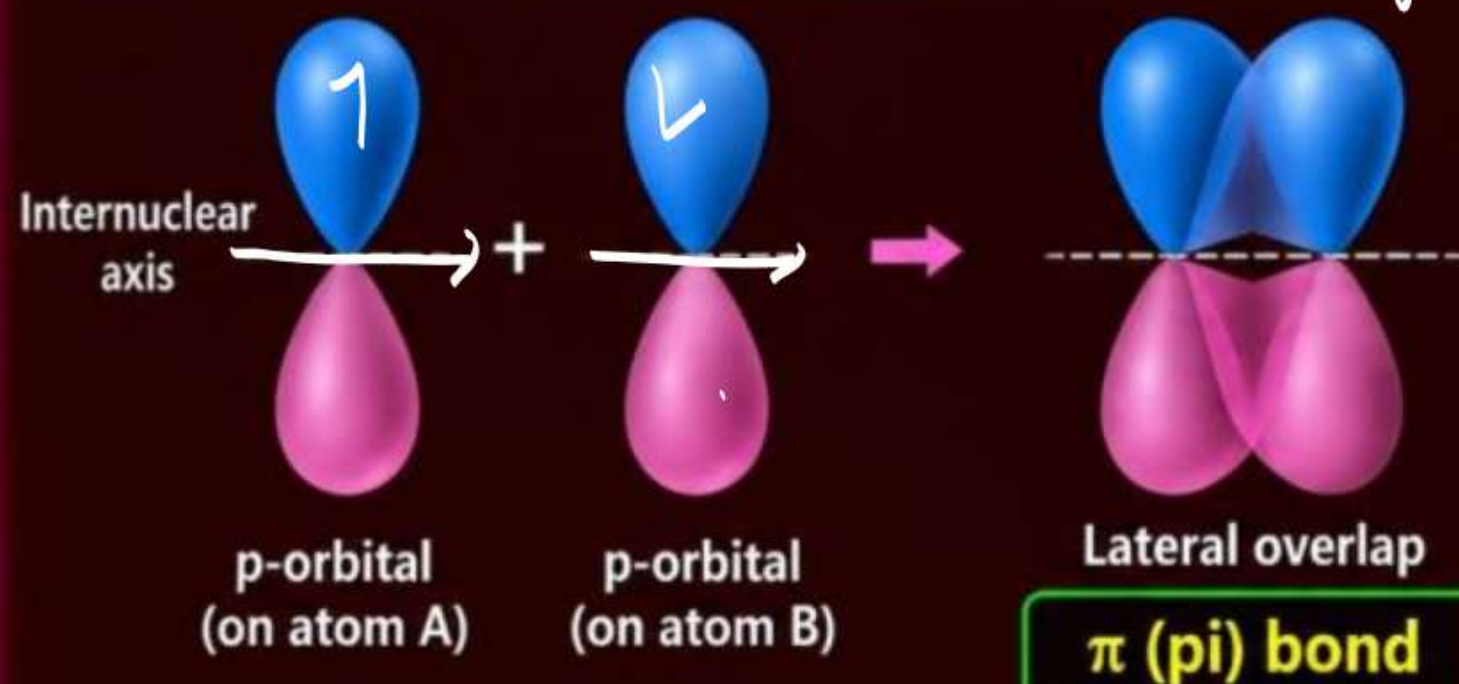
Types of Bond

2. Pi (π) Bond

Formed by **sideways (lateral) overlap** of p-orbitals.

perpendicular to the internuclear axis

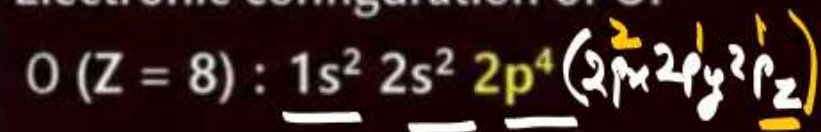
Lateral Overlap of p-Orbitals



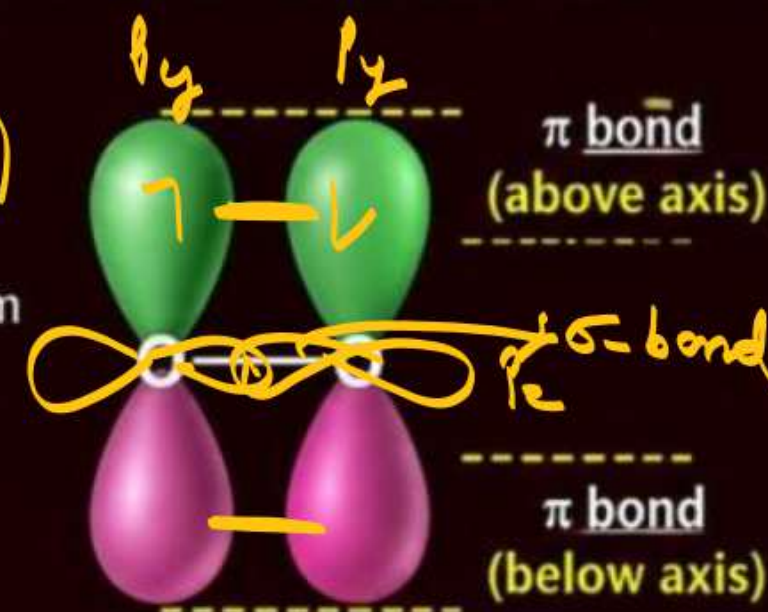
- π bond is formed by **lateral overlap** of two parallel p-orbitals.
- The two orbitals overlap **sidewise** (above and below the internuclear axis).
- Electron density is present above and below the **bond axis**.

Example: Formation of O_2 (Oxygen molecule)

Electronic configuration of O:



In O_2 , one p-orbital from each O atom undergoes lateral overlap to form a π ($2p-2p$) bond.



Key Takeaway: π bond is formed by lateral overlap of p-orbitals and has electron density above and below the internuclear axis.

σ bond is stronger than π bond (Good overlap).



Hybridisation (by Pauling)

(Important)



Definition

- It is the process in which **atomic orbitals of similar energy** mix to form new orbitals of equal energy and shape.
- These new orbitals are called **hybrid orbitals**. *Phenomena is called hybridization.*

Characteristics:

- 1 Number of hybrid orbitals = number of atomic orbitals mixed.
- 2 All hybrid orbitals have **same energy and shape**.
- 3 They form **stronger and more stable bonds**.
- 4 Type of hybridization decides the **(geometry)** of the molecule.

Conditions for Hybridization:

- 1 Only **valence orbitals** take part.
- 2 The **energy difference** between orbitals should be **small**.
- 3 **Promotion of electron** is not necessary for hybridization.



Types of Hybridization (by Pauling)

Type	Atomic orbitals mixed (with specific orbitals)	Number of hybrid orbitals	Geometry (Bond angle)	Shape (3D view)	Example(s)
sp	$1s + 1p (p_z)$ (s, p_z)	2 (sp)	Linear (180°)		BeCl_2 , CO_2
sp ²	one s + two p (p_x, p_y) (s, p_x, p_y)	3 (sp^2)	Trigonal planar (120°)		BF_3 , SO_3 , C_2H_4
sp ³	one s + three p (p_x, p_y, p_z) (s, p_x, p_y, p_z)	4 (sp^3)	Tetrahedral (109.5°)		CH_4 , NH_3 , H_2O
sp ³ d	one s + three p + one d (d_{z^2}) (s, p_x, p_y, p_z, d_{z^2})	5 (sp^3d)	Trigonal bipyramidal ($90^\circ, 120^\circ$)		PCl_5 , SF_4 , BrF_3
sp ³ d ²	one s + three p + two d ($d_{z^2}, d_{x^2-y^2}$) (s, $p_x, p_y, p_z, d_{z^2}, d_{x^2-y^2}$)	6 (sp^3d^2)	Octahedral (90°)		SF_6 , $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$
sp ³ d ³	one s + three p + three d (d_{xy}, d_{yz}, d_{xz}) (s, $p_x, p_y, p_z, d_{xy}, d_{yz}, d_{xz}$)	7 (sp^3d^3)	Pentagonal bipyramidal ($72^\circ, 90^\circ$)		IF_7 , $[\text{ZrF}_7]^{3-}$

Key Takeaway: Hybridization = mixing of **specific orbitals** → decides the **geometry** of the molecule.

Example: BeCl_2 (sp Hybridization)

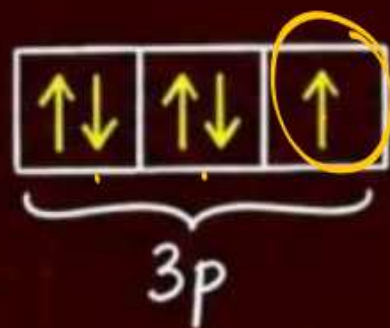
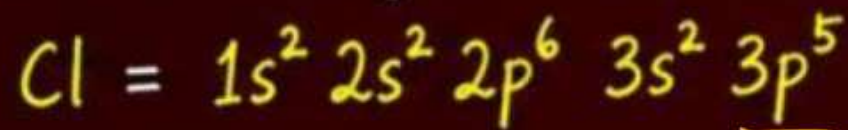
Central atom - Be ($Z=4$)

Electronic configuration of Be :



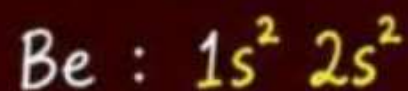
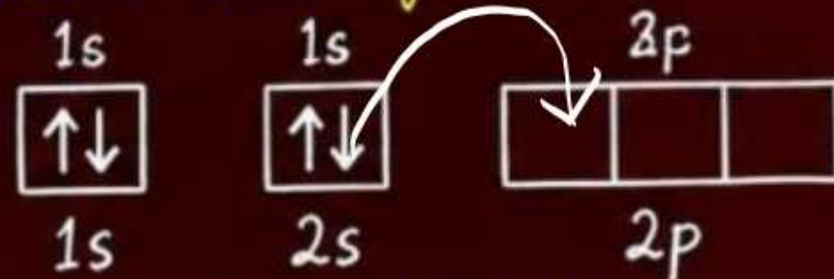
Side atom - Cl ($Z=17$)

Electronic configuration of Cl :

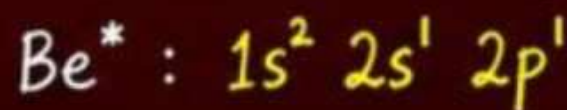
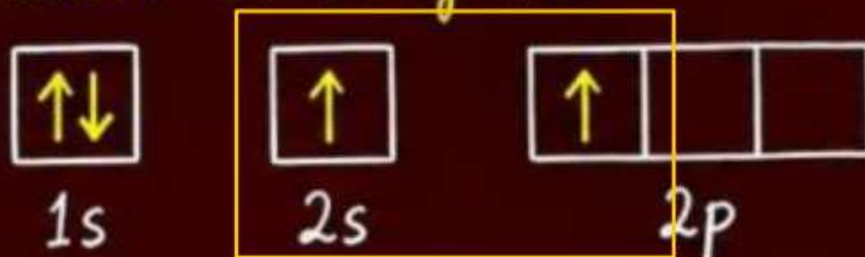


one unpaired
electron for
bonding

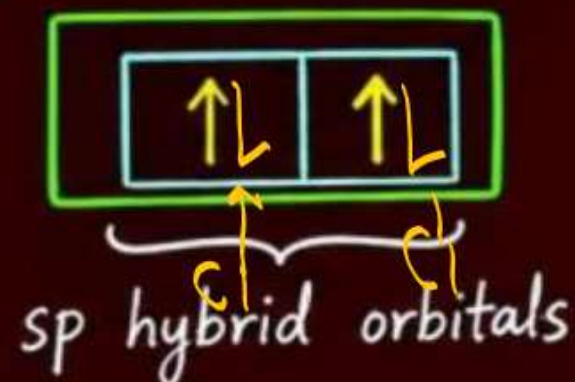
Ground state of Be :



Excited state of Be :



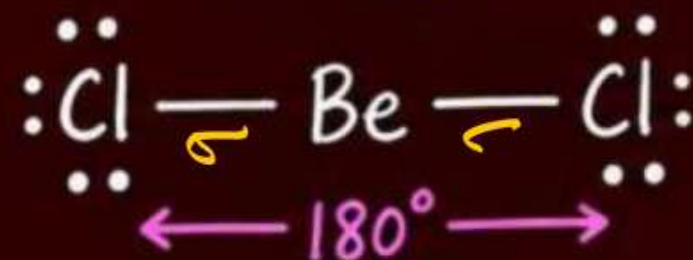
Hybridization of Be :



Each Cl donates
one lone pair to Be .



BeCl_2 :



Thus, BeCl_2 is a linear
molecule with sp hybridization
on Be .

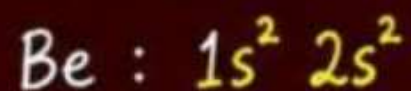
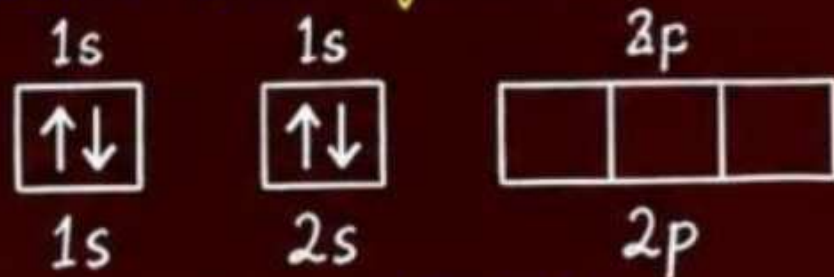
Example: BeCl_2 (sp Hybridization)

Central atom - Be ($Z = 4$)

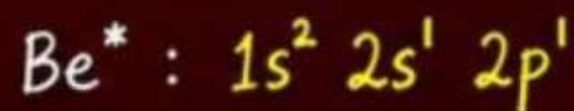
Electronic configuration of Be :



Ground state of Be :



Excited state of Be :



Hybridization of Be :

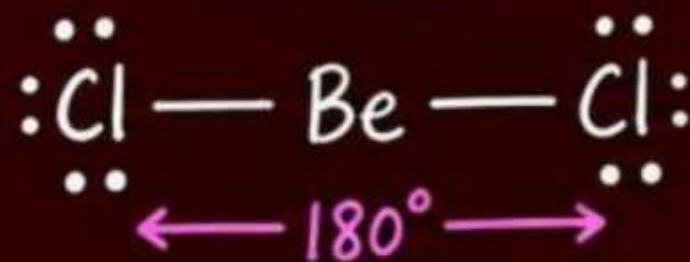


sp hybrid orbitals

Each Cl donates one lone pair to Be.



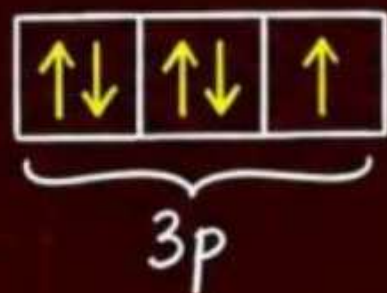
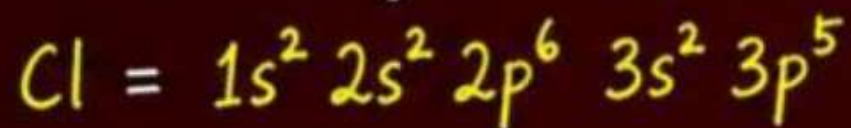
BeCl_2 :



Thus, BeCl_2 is a linear molecule with sp hybridization on Be.

Side atom - Cl ($Z = 17$)

Electronic configuration of Cl :



one unpaired electron for bonding

Example: BF_3 (sp^2)

Central atom = B
($Z = 5$)

Electronic configuration of B:
 $\text{B} = 1s^2 2s^2 2p^1$

Side atom = F
($Z = 9$)

Electronic configuration of F:
 $\text{F} = 1s^2 2s^2 2p^5$
($2p_x^2 2p_y^2 2p_z^1$)

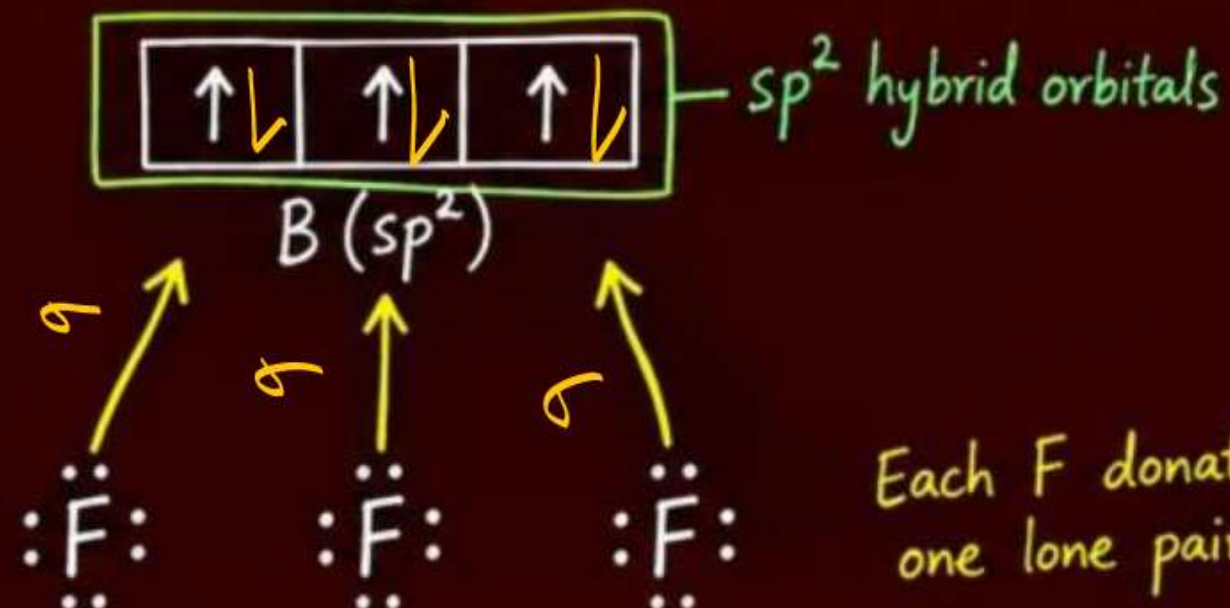
Ground state E.C. of B:



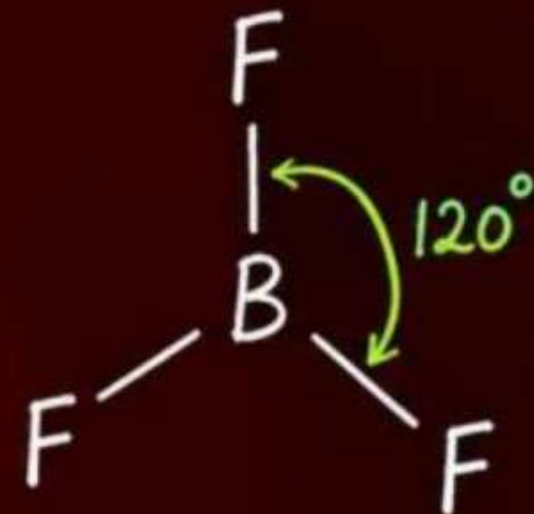
Excited state E.C. of B:



Hybridization (sp^2):



Structure of BF_3 :



Thus, BF_3 is a trigonal planar molecule with sp^2 hybridization on B.

Example: NH_3

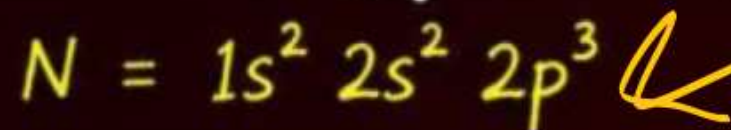
Geometry: Tetrahedral (109.5°)

Molecule: NH_3

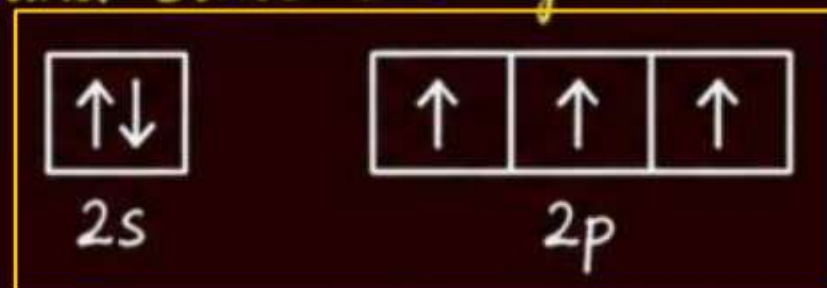
Central atom = Nitrogen (N) ($Z=7$)

Side atom = H ($Z=1$) $\rightarrow 1s^1$

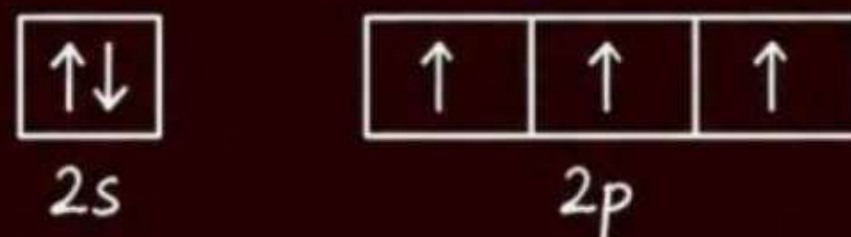
Electronic configuration of N:



Ground state E.C. of N:



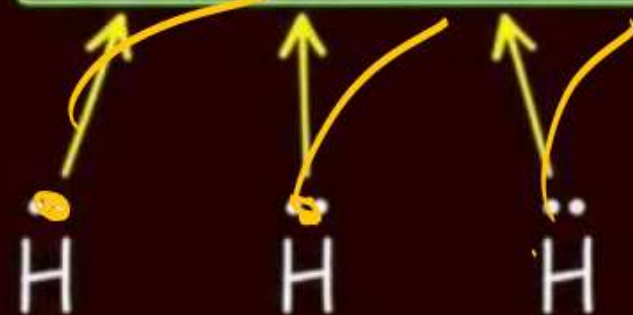
~~Excited state E.C. of N:~~



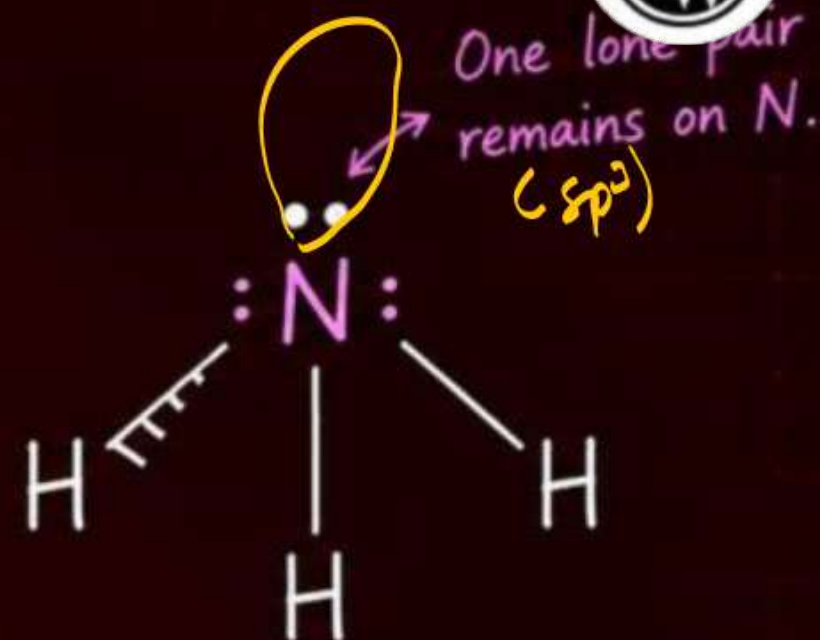
Hybridization (sp^3):



$\leftarrow sp^3$ hybrid orbitals of N



Each H donates one electron to one sp^3 hybrid orbital of N to form an N-H σ bond.



One lone pair remains on N. (sp^3)

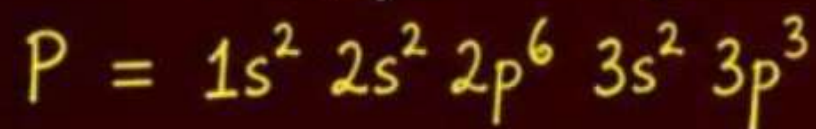
Shape: Trigonal pyramidal (due to one lone pair)

Thus, NH_3 has sp^3 hybridization on N and tetrahedral electron pair geometry (109.5°).

Example: PCl_5 (sp^3d Hybridization)

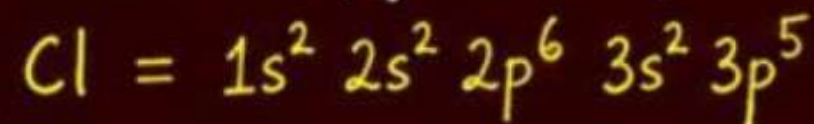
Central atom = P ($Z = 15$)

Electronic configuration of P :

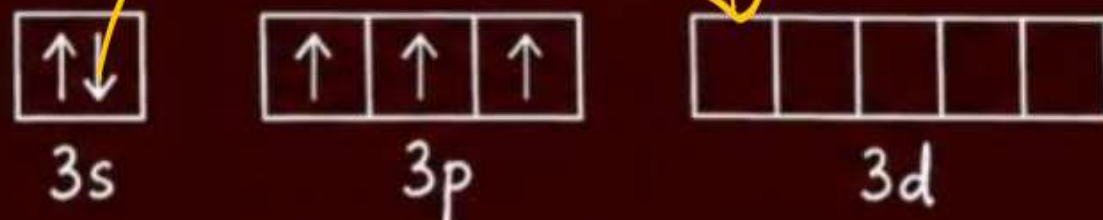


Side atom = Cl ($Z = 17$)

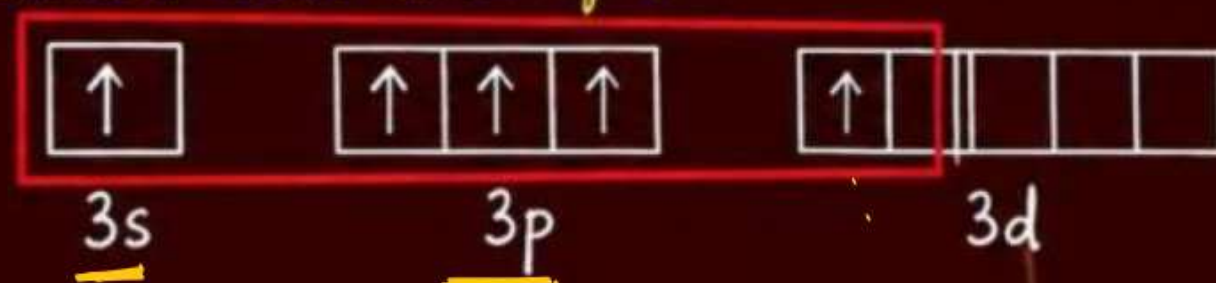
Electronic configuration of Cl :



Ground state E.C. of P :



Excited state E.C. of P :

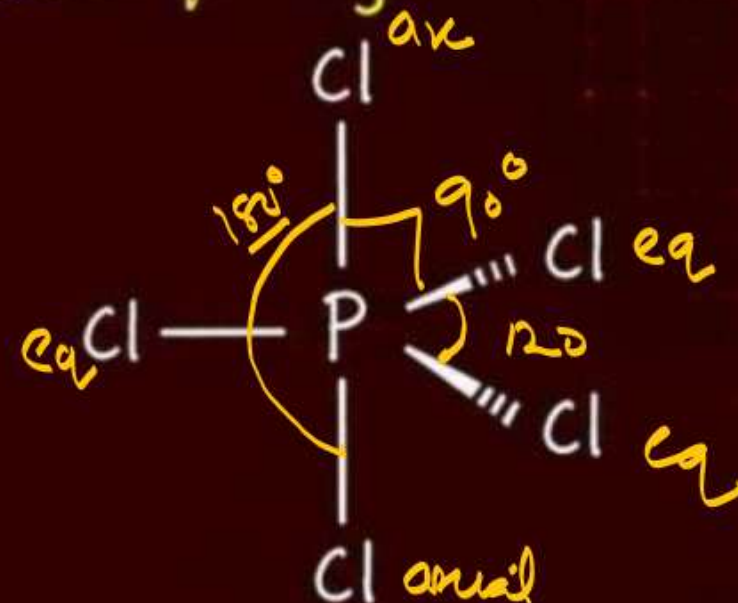


Hybridization: $3s + 3p_x + 3p_y + 3p_z + d_{z^2} \rightarrow 5 sp^3d$ hybrid orbitals



Each Cl donates one lone pair to one sp^3d hybrid orbital of P to form five $\text{P}-\text{Cl}$ σ bonds.

Structure of PCl_5 :



Geometry: Trigonal bipyramidal

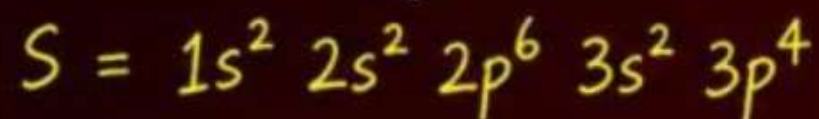
Bond angles: $90^\circ, 120^\circ, 180^\circ$

Thus, PCl_5 has sp^3d hybridization on P with trigonal bipyramidal geometry.

Example: SF_6 (sp^3d^2 Hybridization)

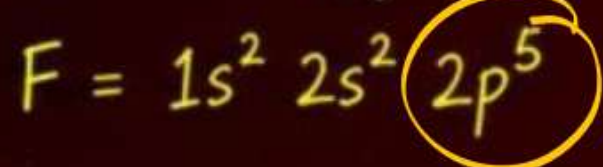
Central atom = S ($Z=16$)

Electronic configuration of S:



Side atom = F ($Z=9$)

Electronic configuration of F:



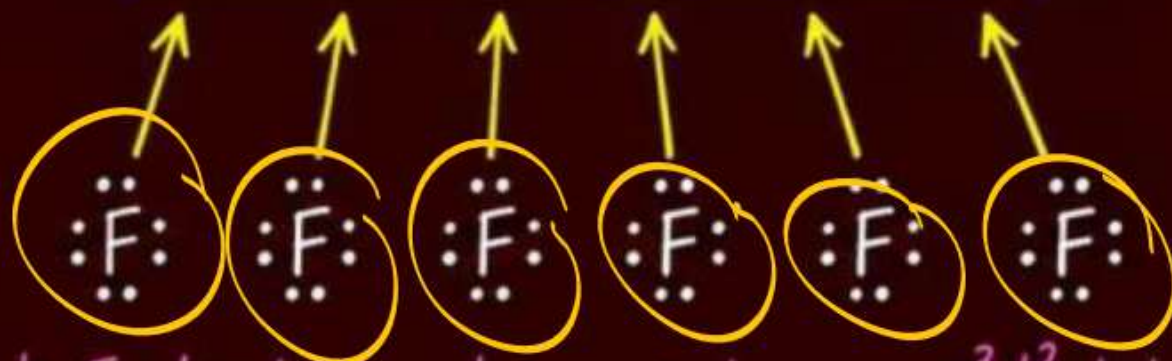
Ground state E.C. of S:



Excited state E.C. of S:



Hybridization: $3s + 3p_x + 3p_y + 3p_z + d_{x^2-y^2} + d_{z^2} \rightarrow 6 sp^3d^2$ hybrid orbitals



Each F donates one lone pair to one sp^3d^2 hybrid orbital of S to form six S-F σ bonds.

Structure of SF_6 :



Geometry: Octahedral

Bond angle: 90°

Thus, SF_6 has sp^3d^2 hybridization on S and octahedral geometry.

Example:



(sp^3d^3)

Ground state E.C. of I:



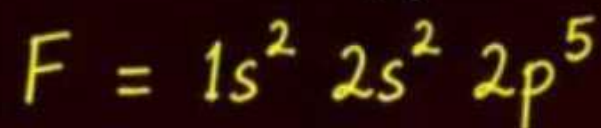
Central atom = I ($Z = 53$)

Side atom = F ($Z = 9$)

Electronic configuration of I:



Electronic configuration of F:

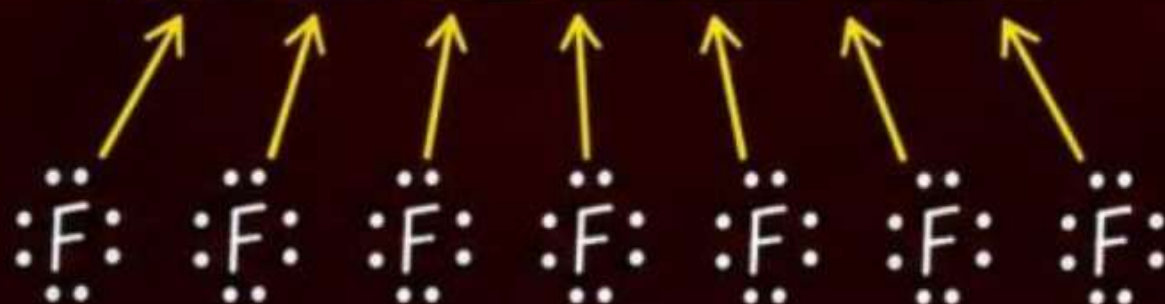


(sp^3)

Excited state E.C. of I:

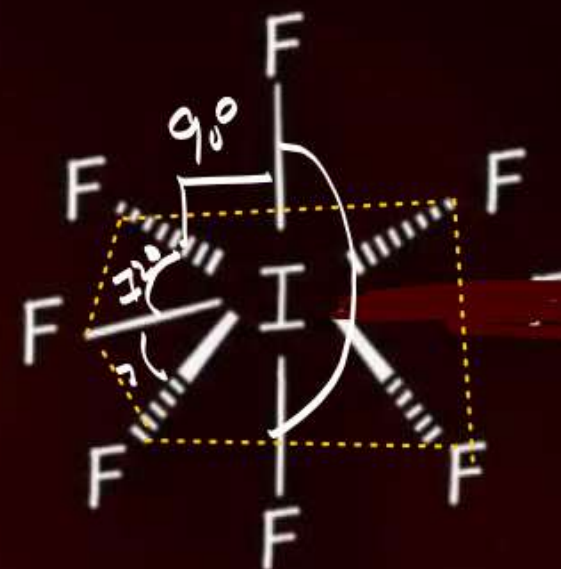


Hybridization: $5s + 5p + 5d \rightarrow 7 sp^3d^3$ hybrid orbitals



Each F donates one lone pair to one sp^3d^3 hybrid orbital of I to form seven I-F σ bonds.

Structure of IF_7 :



Geometry: Pentagonal bipyramidal

Bond angles: 72° , 90° , 180°

Thus, IF_7 has sp^3d^3 hybridization on I and pentagonal bipyramidal geometry.

T.E.P. (Total Electron Pairs)	Type of Hybridization	Geometry
2	sp	Linear
3	sp^2	Trigonal planar
4	sp^3	Tetrahedral
5	sp^3d	Trigonal bipyramidal
6	sp^3d^2	Octahedral
7	sp^3d^3	Pentagonal bipyramidal

1. $\text{SF}_4 \rightarrow$
2. $\text{XeF}_4 \rightarrow$
3. $\text{CO}_3^{2-} \rightarrow$
4. $\text{CO}_4^{2-} \rightarrow$
5. $\text{SO}_3 \rightarrow$
6. $\text{ClF}_3 \rightarrow$
7. $\text{NH}_4^+ \rightarrow$
8. $\text{BCl}_3 \rightarrow$
9. $\text{PCl}_5 \rightarrow$
10. $\text{H}_2\text{O} \rightarrow$

a) Dipole moment
OR
b) Hydrogen bonding.

c) both

d) None.

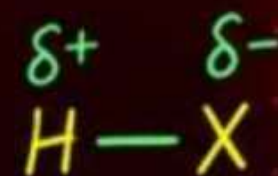
Hydrogen Bonding

Definition :

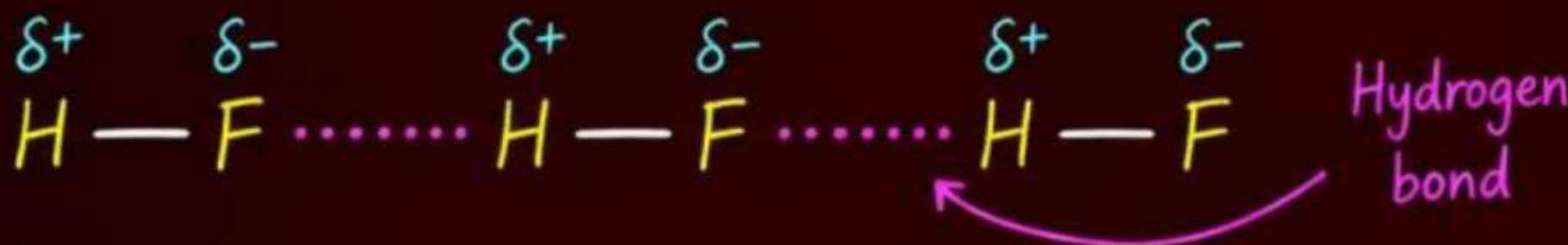
- It is the attractive force between a **hydrogen atom** (bonded to a highly electronegative atom - **F, O or N**) and another electronegative atom of the same or a different molecule.
- Represented by a dotted line (---).

Cause of Formation :

- When H is bonded to **F, O or N**, electrons' shift towards that atom due to its **high electronegativity**.
- Hydrogen becomes **partially positive (δ^+)** and the other atom becomes **partially negative (δ^-)**.
- This leads to **electrostatic attraction** between molecules, forming a **hydrogen bond**.



e.g.



Occurs between molecules of the **same or different** molecules.

Types of Hydrogen Bonds

(a) Intermolecular Hydrogen Bond

Formed between different molecules (same or different compounds).

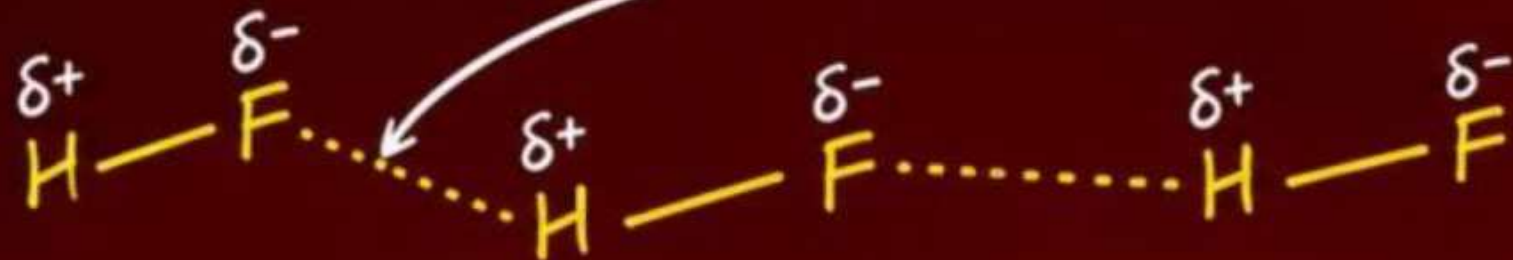
Example: H_2O , HF, alcohols, *p*-nitro phenol

Causes association of molecules, leading to:

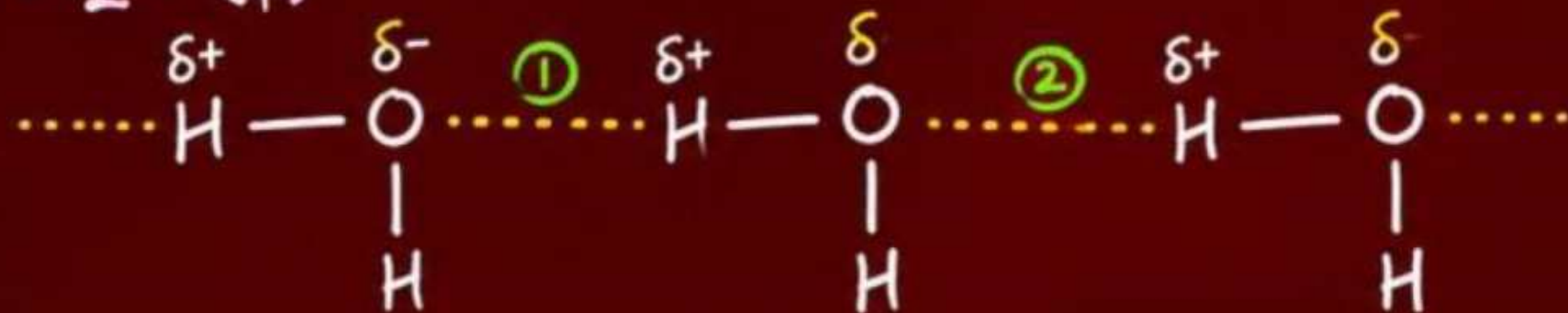
Higher melting point, boiling point, and solubility.



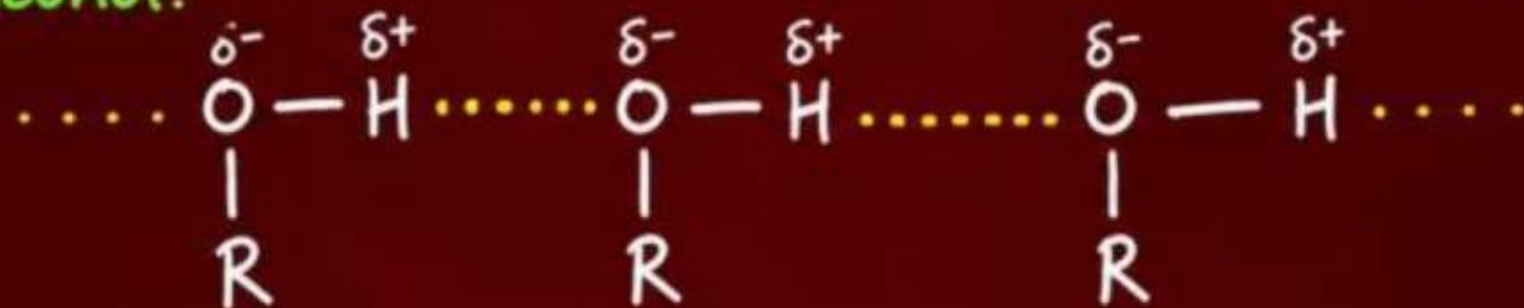
e.g a) $\text{H}-\text{F}$ (solid state) $\rightarrow$ H-bond



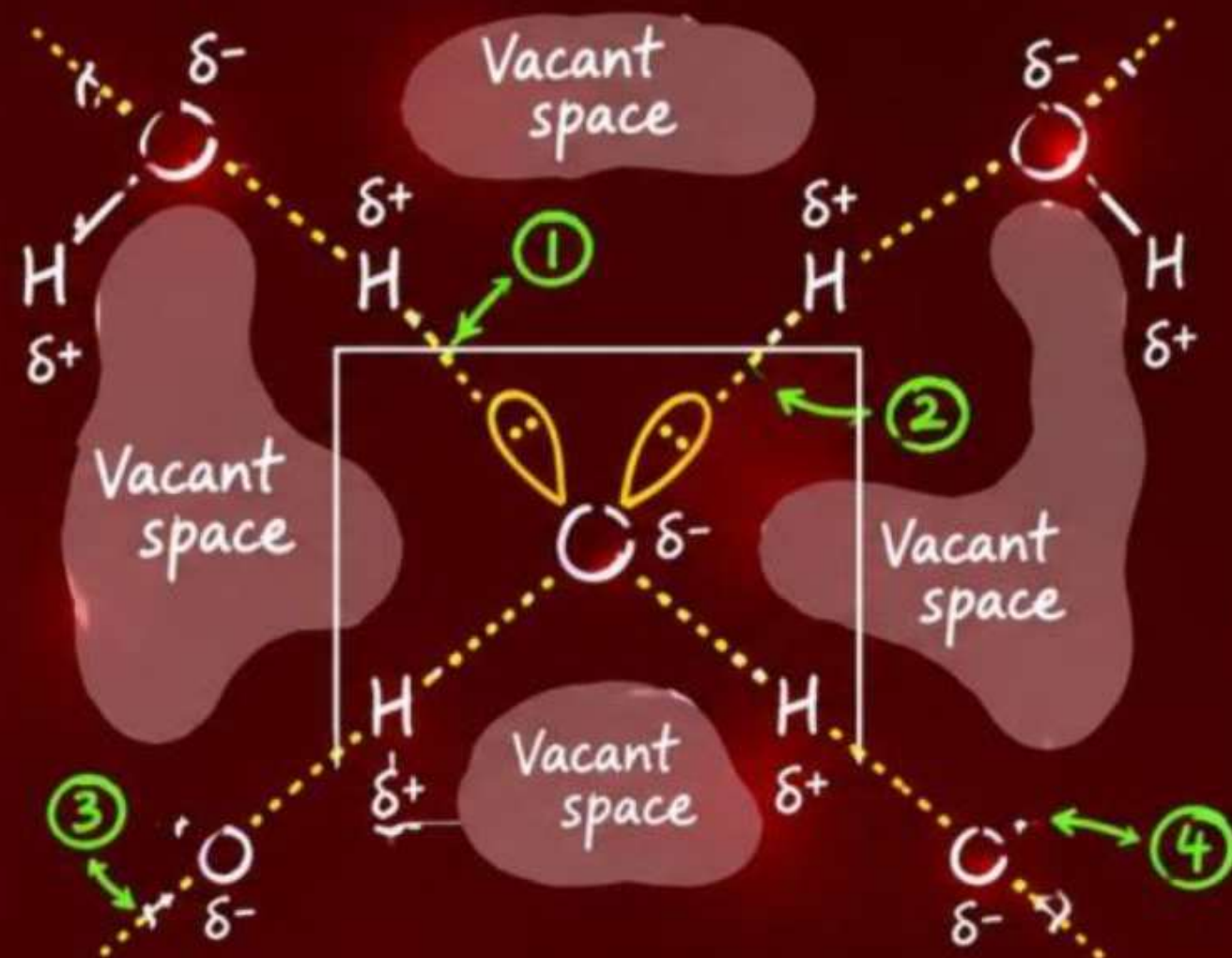
b) H_2O (l)



c) Alcohol.



H_2O (s) (ice).

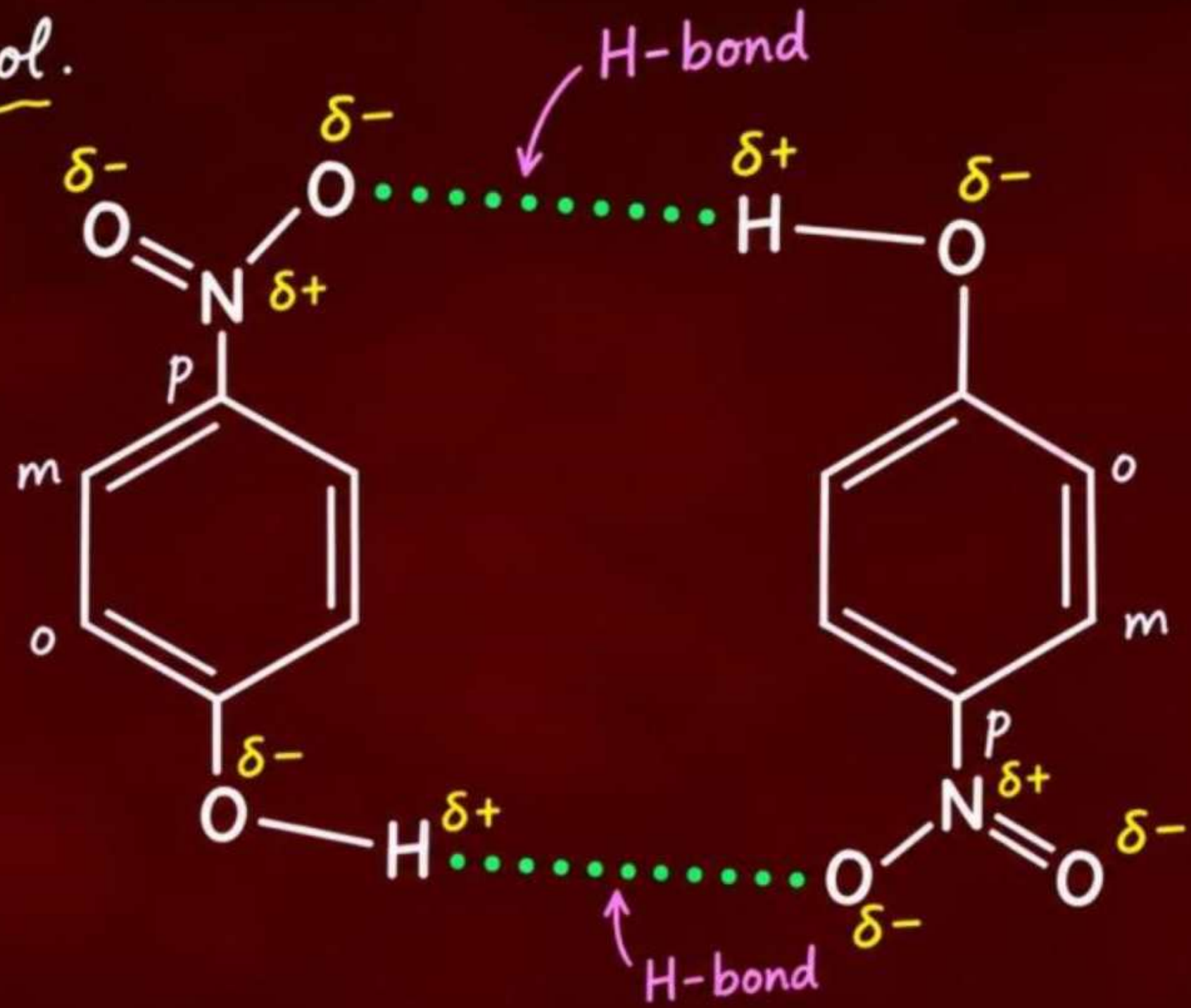


● **Density** : $\rho_{\text{H}_2\text{O}(\text{l})} > \rho_{\text{H}_2\text{O}(\text{s})}$
 $(\approx 1.0 \text{ g mL}^{-1} \quad \approx 0.92 \text{ g mL}^{-1})$

● **Why ice floats on water ?**

- $\rightarrow$ In ice, H_2O molecules are held in an open hexagonal structure due to hydrogen bonding.
- $\rightarrow$ This increases the volume and decreases the density.
- $\rightarrow$ Hence, ice (less dense) floats on water (more dense).

d) p-nitro phenol.



p-Nitro phenol molecules are associated through **intermolecular hydrogen bonding** to form a dimer.

Intramolecular Hydrogen Bond

- When a H atom is attached to a highly electronegative atom (O, N, F) and forms a hydrogen bond with another electronegative atom present in the **same molecule**, it is called **intramolecular hydrogen bonding**.
- This leads to a stable **six-membered chelate ring**.
- It prevents association of molecules, so it **lowers boiling point, melting point and solubility**.



o-Nitrophenol

Example : o-Nitrophenol has lower boiling point than p-nitrophenol due to intramolecular hydrogen bonding.

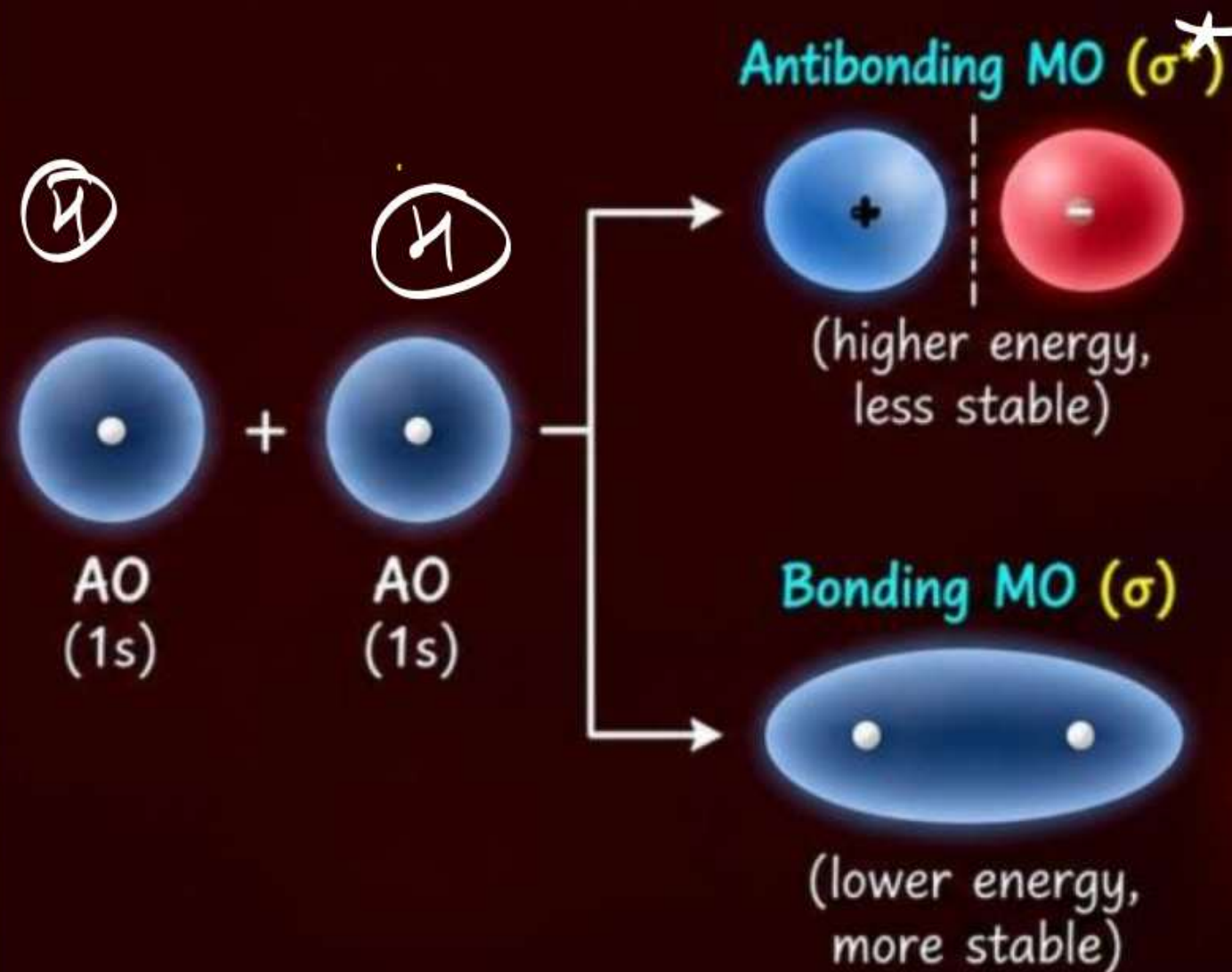
Molecular Orbital Theory

(Most Imp)



1. Molecules have **molecular orbitals (MOs)** with definite energy levels where electrons reside.
2. MOs form by combination of **atomic orbitals (AOs)** of **proper symmetry** and **comparable energy**.
3. AOs are **monocentric** (one nucleus), whereas MOs are **polycentric** (two or more nuclei).
4. Electron probability around one nucleus is described by an **AO**, while around multiple nuclei by an **MO**.
5. Number of MOs formed = number of AOs combined.
e.g. **2 AOs $\rightarrow$ 2 MOs**.
6. These are **bonding MOs** (lower energy, more stable) and **antibonding MOs** (higher energy, less stable).
7. Electrons fill MOs according to **Aufbau principle**, **Pauli exclusion principle** and **Hund's rule**.

Combination of two AOs



Molecular Orbital Diagram (MOD)



- The arrangement of molecular orbitals (MOs) in the order of increasing energy is called the Molecular Orbital Diagram (MOD).

Types of MOD:

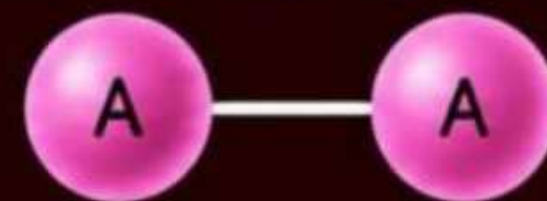
- 1 For homonuclear diatomic molecules from H_2 to N_2 ($Z = 1$ to 7)

Examples : H_2 , Li_2 , Be_2 , B_2 , C_2 , N_2

- 2 For homonuclear diatomic molecules from O_2 to Ne_2 ($Z = 8$ to 10)

Examples : O_2 , F_2 , Ne_2

Homonuclear diatomic molecule



(Same atoms)

e.g. H_2 , O_2 , N_2

- ✓ Due to difference in relative energies of $2s$ and $2p$ orbitals, the order of MOs (and hence MOD) is different for H_2-N_2 and for O_2-Ne_2 .

Molecular Orbital Diagrams for Homonuclear Diatomic Molecules

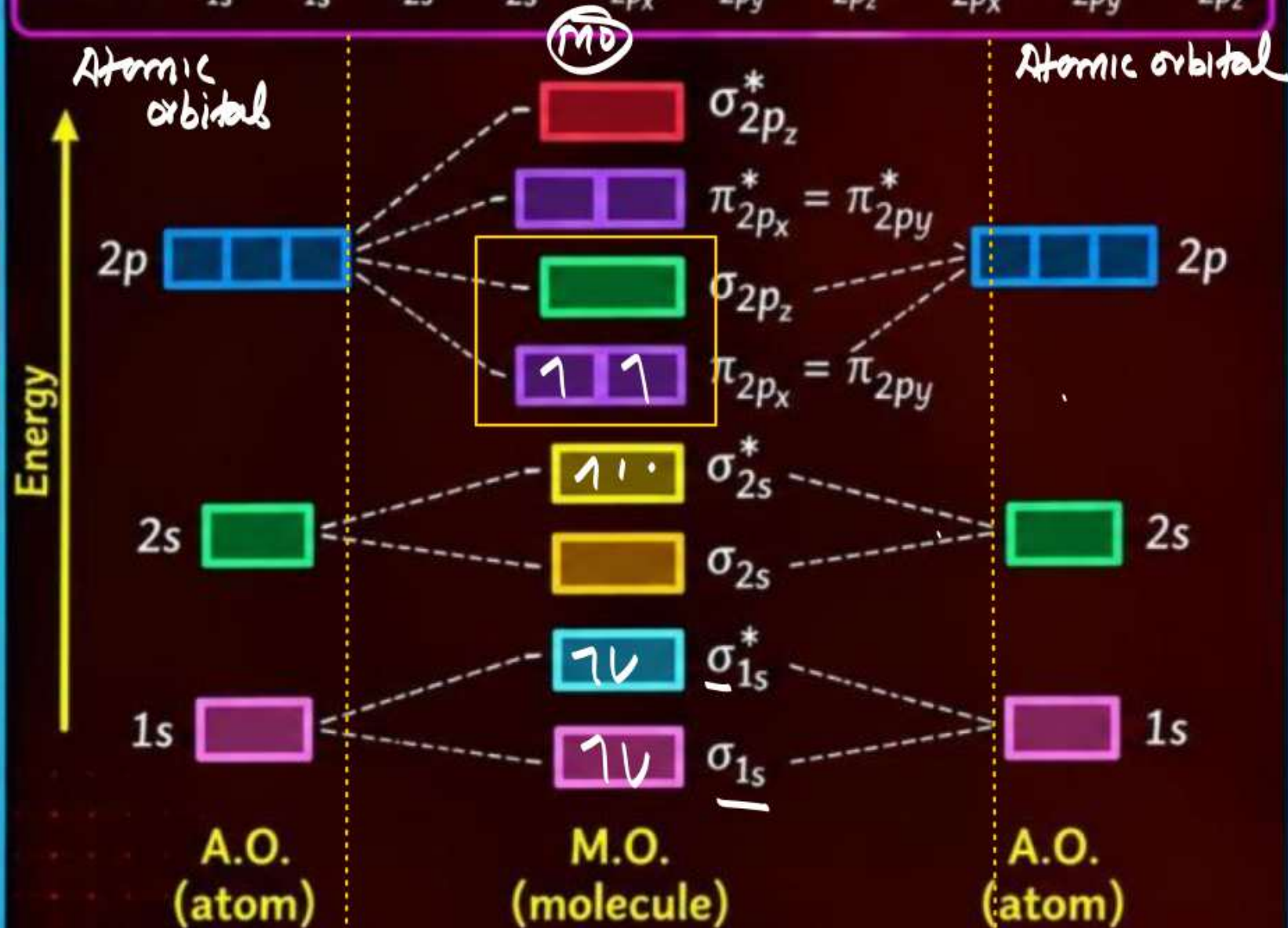


arrangement of MO in increasing order of energy.

1. For H_2 to N_2 ($Z = 1$ to 7)

(e.g. H_2 , Li_2 , Be_2 , B_2 , C_2 , N_2)

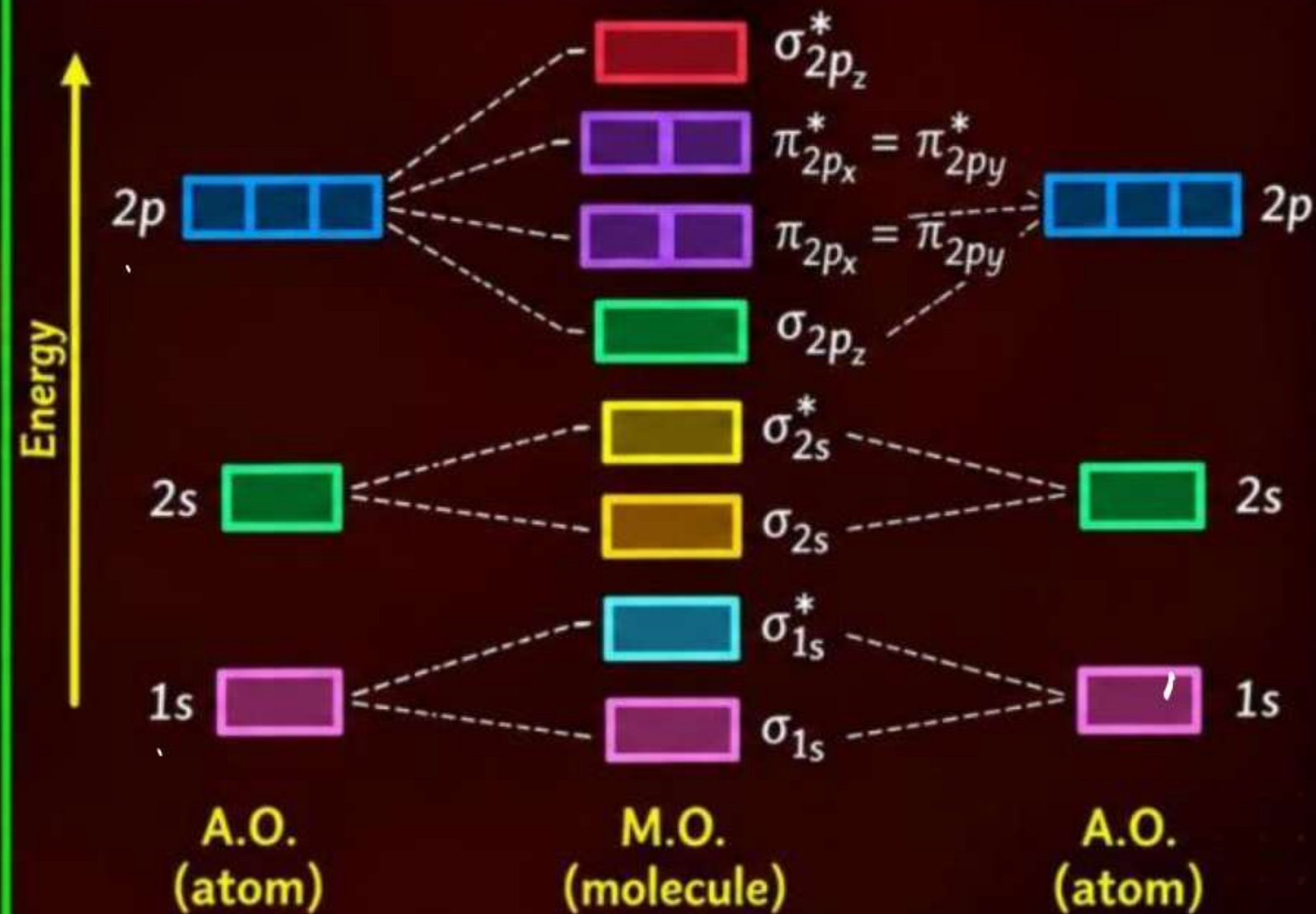
Order: $\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* < \pi_{2p_x} = \pi_{2p_y} < \sigma_{2p_z} < \pi_{2p_x}^* = \pi_{2p_y}^* < \sigma_{2p_z}^*$



2. For O_2 to Ne_2 ($Z = 8$ to 10)

(e.g. O_2 , F_2 , Ne_2)

Order: $\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* < \sigma_{2p_z} < \pi_{2p_x} = \pi_{2p_y} < \pi_{2p_x}^* = \pi_{2p_y}^* < \sigma_{2p_z}^*$



e.g.

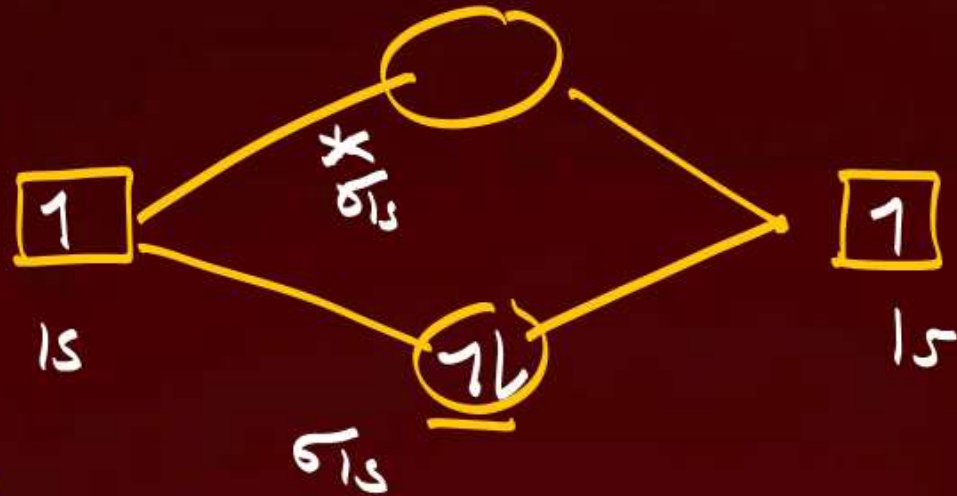
$$H_2 = \underline{\sigma_{1s}^2}$$

$$\text{Total} = 2$$

• Stability $\Rightarrow$ Stable ($N_b > N_a$).

$$BO = \frac{1}{2}(N_b - N_a) = \frac{1}{2}(2 - 0) = \frac{2}{2} = 1$$

• Magnetic $\rightarrow$ Diamagnetic



$$\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* < \pi_{2px} = \pi_{2py} < \sigma_{2pz} \\ < \pi_{2px}^* = \pi_{2py}^* < \sigma_{2pz}^*$$

e.g. B_2 .

Total $\bar{e} = 10$

M.O. $\bar{e}$ configuration $\Rightarrow$

$$\underline{\sigma_{1s}^2} < \underline{\sigma_{1s}^{*2}} < \underline{\sigma_{2s}^2} < \underline{\sigma_{2s}^{*2}} < \underline{\pi_{2px}^2 = \pi_{2py}^2}$$

Stability $\Rightarrow N_b > N_a$ i.e. stable

$$B.O = \frac{1}{2}(N_b - N_a) = \frac{1}{2}(6 - 4) = \frac{1}{2} \times 2 = 1$$

Magnetic behaviour $\Rightarrow$ Paramagnetic.

$$\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* < \pi_{2px} = \pi_{2py} < \sigma_{2pz} < \pi_{2px}^* = \pi_{2py}^* < \sigma_{2pz}^*$$

Molecular Orbital Electronic Configurations (O_2 to Ne_2)



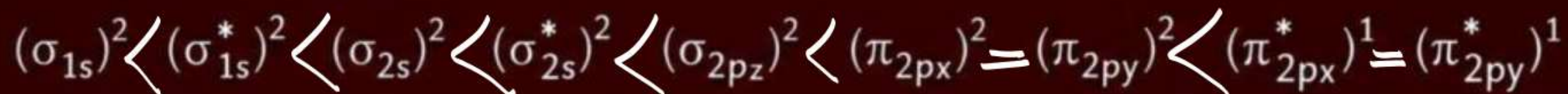
O_2 (Z = 8)

$O = 1s^2 2s^2 2p^4$

Total electrons = 16

Energy order : $\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* < \sigma_{2pz} < \pi_{2px} = \pi_{2py} < \pi_{2px}^* = \pi_{2py}^* < \sigma_{2pz}^*$

MO electronic configuration :

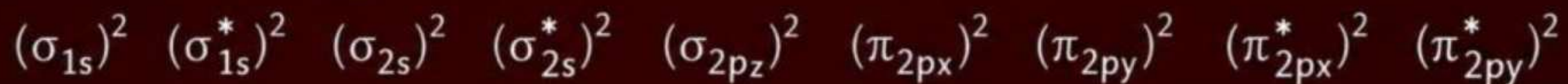


F_2 (Z = 9)

Total electrons = 18

Energy order : $\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* < \sigma_{2pz} < \pi_{2px} = \pi_{2py} < \pi_{2px}^* = \pi_{2py}^* < \sigma_{2pz}^*$

MO electronic configuration :



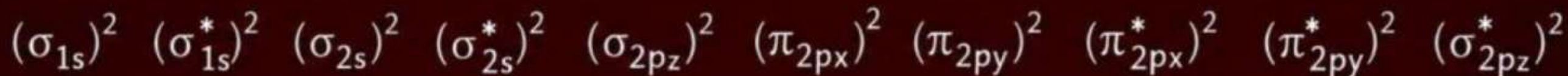
Ne_2 (Z = 10)

$Ne = 1s^2 2s^2 2p^6$

Total electrons = 20

Energy order : $\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* < \sigma_{2pz} < \pi_{2px} = \pi_{2py} < \pi_{2px}^* = \pi_{2py}^* < \sigma_{2pz}^*$

MO electronic configuration :



Molecular Orbital Electronic Configurations (O_2 to Ne_2)

O_2 ($Z = 8$)

Total electrons = 16

MO electronic configuration :

$$(\sigma_{1s})^2 < (\sigma_{1s}^*)^2 < (\sigma_{2s})^2 < (\sigma_{2s}^*)^2 < (\sigma_{2pz})^2 < (\pi_{2px})^2 = (\pi_{2py})^2 < (\pi_{2px}^*)^1 = (\pi_{2py}^*)^1 < (\sigma_{2pz}^*)^0$$

F_2 ($Z = 9$)

Total electrons = 18

MO electronic configuration :

$$(\sigma_{1s})^2 < (\sigma_{1s}^*)^2 < (\sigma_{2s})^2 < (\sigma_{2s}^*)^2 < (\sigma_{2pz})^2 < (\pi_{2px})^2 = (\pi_{2py})^2 < (\pi_{2px}^*)^2 = (\pi_{2py}^*)^2 < (\sigma_{2pz}^*)^0$$

Ne_2 ($Z = 10$)

Total electrons = 20

MO electronic configuration :

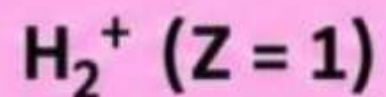
$$(\sigma_{1s})^2 < (\sigma_{1s}^*)^2 < (\sigma_{2s})^2 < (\sigma_{2s}^*)^2 < (\sigma_{2pz})^2 < (\pi_{2px})^2 = (\pi_{2py})^2 < (\pi_{2px}^*)^2 = (\pi_{2py}^*)^2 < (\sigma_{2pz}^*)^2$$

$$\textcircled{Ne_2}$$

$$= \frac{1}{2} (N_b - N_a)$$

$$= \frac{1}{2} (10 - 10) = 0$$

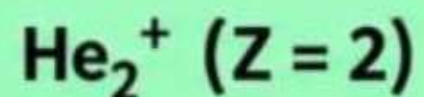
Molecular Orbital Electronic Configurations (Selected Species)



Total electrons = 1

MO electronic configuration :

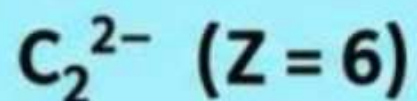
$$(\sigma_{1s})^1 < (\sigma_{1s}^*)^0 < (\sigma_{2s})^0 < (\sigma_{2s}^*)^0 < (\sigma_{2p_z})^0 < (\pi_{2p_x})^0 = (\pi_{2p_y})^0 < (\pi_{2p_x}^*)^0 = (\pi_{2p_y}^*)^0 < (\sigma_{2p_z}^*)^0$$



Total electrons = 3

MO electronic configuration :

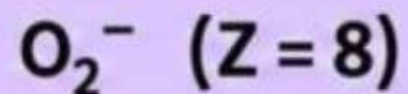
$$(\sigma_{1s})^2 < (\sigma_{1s}^*)^1 < (\sigma_{2s})^0 < (\sigma_{2s}^*)^0 < (\sigma_{2p_z})^0 < (\pi_{2p_x})^0 = (\pi_{2p_y})^0 < (\pi_{2p_x}^*)^0 = (\pi_{2p_y}^*)^0 < (\sigma_{2p_z}^*)^0$$



Total electrons = 10

MO electronic configuration :

$$(\sigma_{1s})^2 < (\sigma_{1s}^*)^2 < (\sigma_{2s})^2 < (\sigma_{2s}^*)^2 < (\pi_{2p_x})^2 = (\pi_{2p_y})^2 < (\sigma_{2p_z})^0 < (\pi_{2p_x}^*)^0 = (\pi_{2p_y}^*)^0 < (\sigma_{2p_z}^*)^0$$



Total electrons = 17

MO electronic configuration :

$$(\sigma_{1s})^2 < (\sigma_{1s}^*)^2 < (\sigma_{2s})^2 < (\sigma_{2s}^*)^2 < (\sigma_{2p_z})^2 < (\pi_{2p_x})^2 = (\pi_{2p_y})^2 < (\pi_{2p_x}^*)^2 = (\pi_{2p_y}^*)^1 < (\sigma_{2p_z}^*)^0$$

Electronic Configuration and Molecular Behaviour

1. Stability

- Molecule is stable if bonding electrons (N_b) > antibonding electrons (N_a).
- Unstable if $N_b \leq N_a$.

2. Bond Order

- Formula : $BO = \frac{1}{2} (N_b - N_a)$
- If $BO > 0 \rightarrow$ stable, $BO = 0 \rightarrow$ unstable.

3. Nature of Bond

- $BO = 1 \rightarrow$ single bond
- $BO = 2 \rightarrow$ double bond
- $BO = 3 \rightarrow$ triple bond

4. Bond Length

- Higher bond order $\rightarrow$ shorter bond length.

5. Magnetic Nature

- All paired electrons $\rightarrow$ **diamagnetic** (not attracted).
- Unpaired electrons $\rightarrow$ **paramagnetic** (attracted).

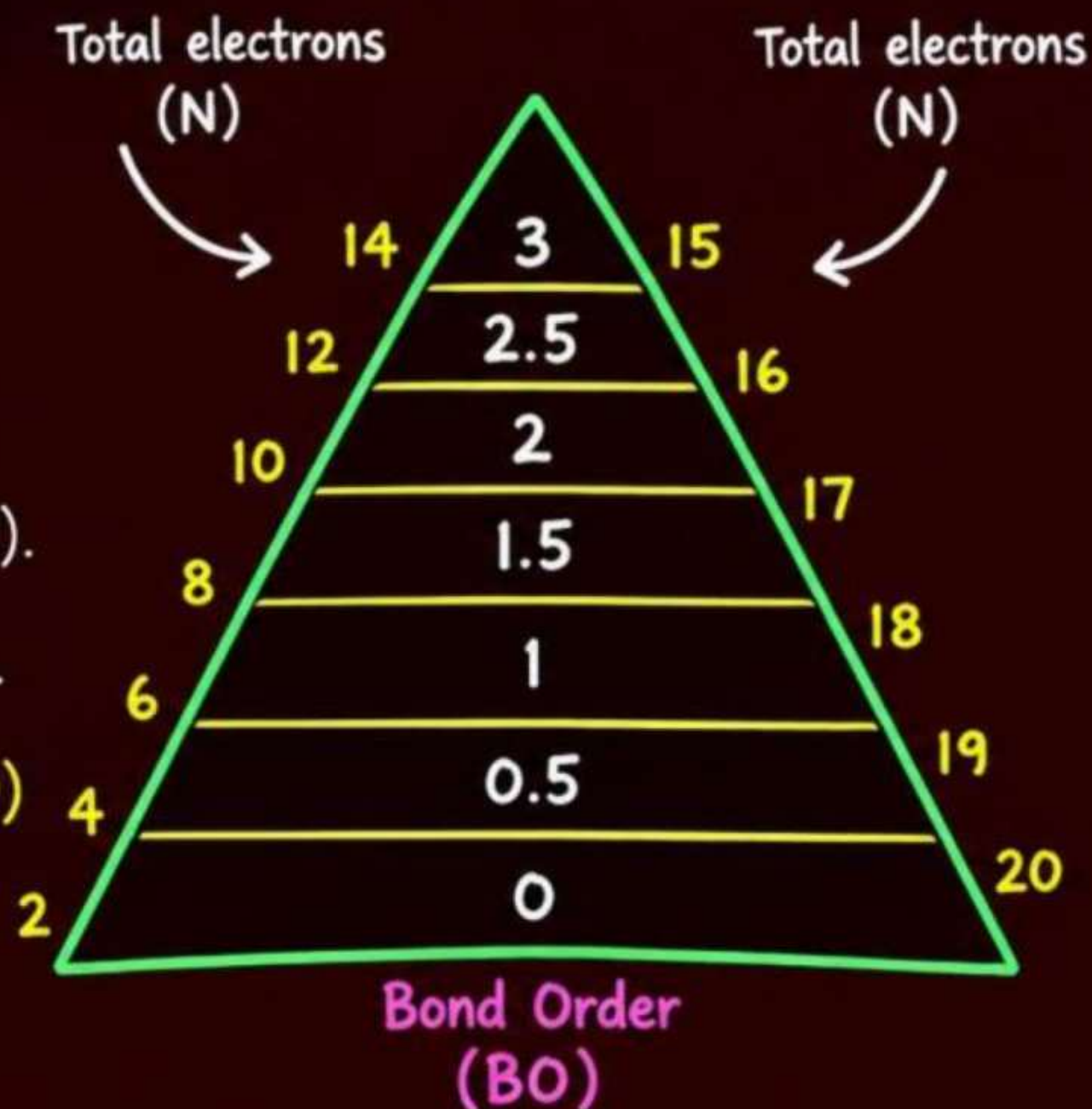
≡ TRICKS ≡



Electronic Configuration and Molecular Behaviour

For determining Bond Order :

1. Count total number of electrons (**N**) in the species (including charge).
2. Locate **N** in the diagram.
3. Read the bond order (**BO**) corresponding to that N.



For Magnetic Properties :

1. If total number of electrons (**N**) is even
→ **Diamagnetic**
(except for $10e^-$ and $16e^-$)
2. If total number of electrons (**N**) is odd
→ **Paramagnetic**

Thank
You

Dipole
moment
&
H-bonding

✓
Youtube